

DATA GOVERNANCE V1

Section 1 – Data Governance Charter & Governance Framework

Document ID: GOV-DG-001

Document Name: Data Governance V1

Section ID: DG-01

Section Title: Data Governance Charter & Governance Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

The purpose of this Data Governance Charter is to establish the governance framework, decision authority structure, accountability model, and operating principles governing data assets within the Transparent Opportunity Decision Platform (TODP).

This charter defines how data is managed, controlled, protected, audited, retained, and governed throughout its lifecycle to ensure transparency, explainability, traceability, auditability, replayability, and enterprise readiness across all platform operations.

The charter serves as the authoritative governance foundation for all data-related activities executed within TODP.

2. Objectives

The objectives of the Data Governance Framework are to:

- Establish enterprise-wide accountability for data ownership and stewardship.
- Define governance responsibilities across all TODP modules.
- Ensure consistency, accuracy, completeness, and reliability of data assets.
- Support regulatory compliance and audit readiness.
- Enable transparent decision-making through governed data lineage.
- Maintain evidence integrity for decision support operations.
- Ensure reproducibility of recommendations through replay-capable datasets.
- Protect sensitive information through controlled access mechanisms.
- Support future enterprise scaling and operational maturity.
- Provide traceable governance controls for internal and external audits.

3. Scope

This governance framework applies to all data assets generated, processed, stored, transmitted, retrieved, analyzed, ranked, recommended, audited, or replayed within TODP.

Governed data domains include:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- User Activity Data

- Operational Metadata
- System Configuration Data
- Traceability Data
- Monitoring Data
- Security Event Data

The framework applies across:

- Application Services
- APIs
- Databases
- Search Indexes
- Object Storage
- Audit Storage
- Backup Repositories
- Analytics Platforms
- Reporting Systems
- Administrative Interfaces

4. Governance Principles

TODP Data Governance shall operate under the following foundational principles:

4.1 Transparency

All data-driven decisions shall be supported by traceable evidence and explainable processing logic.

4.2 Explainability

All recommendation outputs shall be supported by identifiable data sources and reasoning records.

4.3 Traceability

All critical data transformations shall maintain lineage records linking source inputs to downstream outputs.

4.4 Auditability

Governance controls shall generate auditable records sufficient for independent verification.

4.5 Replayability

Decision outcomes shall be reproducible using preserved governance-approved datasets and processing records.

4.6 Accountability

Every governed data asset shall have a designated owner and steward.

4.7 Security

Data access shall follow least-privilege and role-based control principles.

4.8 Compliance

Data processing activities shall comply with applicable legal, contractual, and regulatory obligations.

5. Governance Framework Structure

The TODP Data Governance Framework consists of the following governance layers:

Layer 1 – Strategic Governance

Responsibilities:

- Policy approval
- Governance oversight
- Risk acceptance
- Regulatory alignment

Layer 2 – Operational Governance

Responsibilities:

- Data stewardship
- Quality management
- Access reviews
- Lifecycle governance

Layer 3 – Technical Governance

Responsibilities:

- Data architecture controls
- Metadata management

- Lineage implementation
- Security enforcement

Layer 4 – Assurance Governance

Responsibilities:

- Audit execution
- Compliance verification
- Control effectiveness reviews
- Governance reporting

6. Governance Roles

6.1 Executive Governance Sponsor

Responsibilities:

- Governance program sponsorship
- Strategic oversight
- Resource allocation
- Escalation authority

6.2 Governance Committee

Responsibilities:

- Policy approval
- Governance reviews
- Exception approvals

- Risk governance

6.3 Data Governance Lead

Responsibilities:

- Framework ownership
- Governance operations
- Policy maintenance
- Governance reporting

6.4 Data Owners

Responsibilities:

- Domain accountability
- Data classification approval
- Quality accountability
- Access authorization approval

6.5 Data Stewards

Responsibilities:

- Data quality management
- Metadata maintenance
- Governance control execution
- Operational monitoring

6.6 Security Team

Responsibilities:

- Data protection controls
- Access governance
- Security monitoring
- Incident response

6.7 Audit Team

Responsibilities:

- Governance verification
- Control assessments
- Compliance reviews
- Audit reporting

7. Governance Domains

The framework governs the following TODP domains:

Domain DG-OPP

Opportunity Governance

Domain DG-AUTH

Authority Governance

Domain DG-EVD

Evidence Governance

Domain DG-RET

Retrieval Governance

Domain DG-RANK

Ranking Governance

Domain DG-REC

Recommendation Governance

Domain DG-GOV

Governance Operations

Domain DG-AUD

Audit Governance

Domain DG-REP

Replay Governance

8. Governance Decision Authority

Governance decisions shall be assigned according to authority levels:

Level 1

Operational Decisions

Owner: Data Steward

Level 2

Domain Decisions

Owner: Data Owner

Level 3

Cross-Domain Decisions

Owner: Governance Committee

Level 4

Enterprise Risk Decisions

Owner: Executive Governance Sponsor

All decisions must be recorded within governance audit repositories.

9. Governance Requirements

DG-REQ-001

All governed datasets must have assigned ownership.

DG-REQ-002

All governed datasets must maintain metadata records.

DG-REQ-003

All critical data assets must maintain lineage records.

DG-REQ-004

All access rights must be governed through approved authorization processes.

DG-REQ-005

All governance exceptions must be documented and approved.

DG-REQ-006

All governance controls must be auditable.

DG-REQ-007

All regulated data must be classified before use.

DG-REQ-008

All replay-critical data must be retained according to retention policies.

10. Architectural Constraints

AC-DG-001

Governance controls shall not alter business logic outcomes.

AC-DG-002

Governance mechanisms shall preserve audit evidence integrity.

AC-DG-003

Lineage records shall remain immutable after approval.

AC-DG-004

Replay records shall remain independently verifiable.

AC-DG-005

Governance services shall integrate with Audit and Replay modules.

AC-DG-006

All governance metadata shall support future enterprise scalability.

11. Governance Metrics

The following governance metrics shall be monitored:

DG-MET-001

Data Ownership Coverage %

Target: $\geq 100\%$

DG-MET-002

Metadata Completeness %

Target: $\geq 98\%$

DG-MET-003

Lineage Coverage %

Target: $\geq 95\%$

DG-MET-004

Governance Review Completion %

Target: $\geq 95\%$

DG-MET-005

Access Certification Completion %

Target: $\geq 95\%$

DG-MET-006

Audit Finding Closure %

Target: $\geq 90\%$

12. Audit Requirements

AUD-DG-001

Governance decisions must be auditable.

AUD-DG-002

Data ownership assignments must be verifiable.

AUD-DG-003

Lineage records must support independent validation.

AUD-DG-004

Policy exceptions must maintain approval evidence.

AUD-DG-005

Governance reports must be retained according to retention requirements.

AUD-DG-006

Governance control effectiveness shall be reviewed periodically.

13. Compliance Requirements

The framework supports alignment with:

- SOC 2 Trust Services Criteria
- ISO 27001 Governance Controls
- ISO 27002 Data Governance Controls
- Enterprise Procurement Reviews
- Internal Audit Requirements
- Customer Security Assessments
- Contractual Data Governance Obligations

Compliance alignment shall be validated through periodic governance assessments.

14. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

15. Section Summary

This section establishes the foundational Data Governance Charter and Governance Framework for TODP. It defines governance principles, accountability structures, governance domains, decision authority models, architectural constraints, metrics, audit obligations, and compliance requirements necessary to ensure transparent, explainable, traceable, auditable, and replayable management of all TODP data assets throughout their lifecycle.

DATA GOVERNANCE V1

Section 2 – Data Ownership, Stewardship & Accountability Model

Document ID: GOV-DG-002

Document Name: Data Governance V1

Section ID: DG-02

Section Title: Data Ownership, Stewardship & Accountability Model

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section defines the ownership, stewardship, accountability, and responsibility structure governing all data assets within the Transparent Opportunity Decision Platform (TODP).

The objective of this model is to establish clear accountability for data quality, data protection, access authorization, lifecycle management, compliance adherence, audit readiness, and operational governance across all TODP modules.

The model ensures that every governed dataset has a designated owner, steward, and accountable authority throughout its lifecycle.

2. Objectives

The objectives of the Data Ownership and Stewardship Model are to:

- Establish clear accountability for all governed data assets.

- Eliminate ambiguity regarding data-related decision authority.
- Define responsibilities for data quality and integrity.
- Support governance control execution.
- Ensure consistent regulatory and compliance adherence.
- Strengthen audit readiness.
- Enable effective issue escalation and resolution.
- Support traceability and lineage governance.
- Facilitate enterprise-scale governance operations.
- Maintain accountability across all TODP domains.

3. Scope

This accountability framework applies to:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- User Data
- Metadata Repositories
- Operational Data
- Security Data

- Monitoring Data
- Configuration Data

The model applies to:

- Business Users
- Platform Administrators
- Governance Teams
- Security Teams
- Audit Teams
- Engineering Teams
- Data Consumers
- Third-Party Integrations

4. Governance Accountability Principles

4.1 Single Ownership Principle

Every governed data asset shall have one designated Data Owner.

Shared ownership is prohibited.

4.2 Stewardship Principle

Operational governance activities may be delegated to Data Stewards while accountability remains with Data Owners.

4.3 Accountability Principle

Responsibility may be delegated.

Accountability may not be delegated.

4.4 Least Authority Principle

Governance authority shall be granted only to personnel requiring such authority to fulfill approved responsibilities.

4.5 Auditability Principle

Ownership assignments and accountability changes shall be auditable.

4.6 Traceability Principle

Ownership history shall be preserved throughout the lifecycle of governed data assets.

5. Governance Role Definitions

5.1 Executive Governance Sponsor

Role ID: DG-ROLE-001

Responsibilities:

- Governance sponsorship
- Escalation authority
- Risk acceptance approval

- Strategic governance oversight
- Executive reporting

Accountable For:

- Enterprise governance effectiveness
- Governance funding
- Governance program maturity

5.2 Governance Committee

Role ID: DG-ROLE-002

Responsibilities:

- Governance policy approval
- Governance exception review
- Cross-domain conflict resolution
- Governance performance oversight

Accountable For:

- Governance framework compliance
- Enterprise governance consistency

5.3 Data Governance Lead

Role ID: DG-ROLE-003

Responsibilities:

- Governance program administration
- Governance reporting
- Stewardship coordination
- Governance control monitoring
- Framework maintenance

Accountable For:

- Governance operational effectiveness
- Governance process compliance

5.4 Data Owner

Role ID: DG-ROLE-004

Definition:

The designated business authority accountable for a governed data domain.

Responsibilities:

- Data classification approval
- Access approval authority

- Quality accountability
- Retention approval
- Governance compliance oversight
- Data usage approval

Accountable For:

- Accuracy
- Completeness
- Compliance
- Business relevance
- Data lifecycle decisions

5.5 Data Steward

Role ID: DG-ROLE-005

Definition:

Operational custodian responsible for governance execution.

Responsibilities:

- Metadata management
- Quality monitoring
- Governance control execution
- Data issue management

- Stewardship reporting
- Lineage verification

Accountable For:

- Operational governance activities
- Governance process execution

5.6 Security Owner

Role ID: DG-ROLE-006

Responsibilities:

- Data protection controls
- Security governance
- Access control enforcement
- Incident management
- Security monitoring

Accountable For:

- Confidentiality
- Integrity
- Availability protections

5.7 Audit Owner

Role ID: DG-ROLE-007

Responsibilities:

- Governance audits
- Control testing
- Evidence collection
- Compliance verification

Accountable For:

- Audit quality
- Audit completeness
- Independent verification

6. Data Domain Ownership Structure

6.1 Opportunity Domain

Domain ID: DG-DOM-OPP

Data Owner:

Head of Opportunity Governance

Data Steward:

Opportunity Governance Steward

Primary Accountability:

- Opportunity records
- Opportunity attributes
- Opportunity lifecycle metadata

6.2 Authority Domain

Domain ID: DG-DOM-AUTH

Data Owner:

Authority Governance Lead

Data Steward:

Authority Steward

Primary Accountability:

- Authority sources
- Authority scoring inputs
- Authority lineage

6.3 Evidence Domain

Domain ID: DG-DOM-EVD

Data Owner:

Evidence Governance Lead

Data Steward:

Evidence Steward

Primary Accountability:

- Evidence repositories
- Evidence metadata
- Evidence quality

6.4 Retrieval Domain

Domain ID: DG-DOM-RET

Data Owner:

Retrieval Operations Lead

Data Steward:

Retrieval Steward

Primary Accountability:

- Retrieval logs
- Retrieval metadata
- Retrieval performance records

6.5 Ranking Domain

Domain ID: DG-DOM-RANK

Data Owner:

Ranking Governance Lead

Data Steward:

Ranking Steward

Primary Accountability:

- Ranking inputs
- Ranking outputs
- Ranking lineage

6.6 Recommendation Domain

Domain ID: DG-DOM-REC

Data Owner:

Recommendation Governance Lead

Data Steward:

Recommendation Steward

Primary Accountability:

- Recommendation records
- Recommendation explanations
- Recommendation outcomes

6.7 Governance Domain

Domain ID: DG-DOM-GOV

Data Owner:

Governance Office

Data Steward:

Governance Steward

Primary Accountability:

- Governance decisions
- Policy records
- Governance exceptions

6.8 Audit Domain

Domain ID: DG-DOM-AUD

Data Owner:

Audit Office

Data Steward:

Audit Steward

Primary Accountability:

- Audit evidence
- Audit findings
- Audit reports

6.9 Replay Domain

Domain ID: DG-DOM-REP

Data Owner:

Replay Governance Lead

Data Steward:

Replay Steward

Primary Accountability:

- Replay datasets
- Replay lineage
- Replay verification records

7. Ownership Assignment Requirements

DG-OWN-001

All governed datasets shall have an assigned Data Owner.

DG-OWN-002

All governed datasets shall have an assigned Data Steward.

DG-OWN-003

Ownership assignments shall be recorded in governance repositories.

DG-OWN-004

Ownership changes require documented approval.

DG-OWN-005

Ownership changes shall be logged for audit purposes.

DG-OWN-006

Unassigned datasets are prohibited.

DG-OWN-007

Ownership reviews shall occur at least annually.

8. Stewardship Responsibilities

Data Stewards shall perform:

- Metadata maintenance
- Quality monitoring
- Classification validation
- Issue remediation coordination
- Lineage verification
- Governance reporting
- Retention validation
- Compliance support

Stewards may not:

- Override approved governance policies
- Change classifications without approval
- Grant unauthorized access
- Delete governed audit records

9. Accountability Escalation Model

Level 1

Data Steward

Issue Types:

- Metadata issues
- Data quality issues
- Documentation issues

Level 2

Data Owner

Issue Types:

- Classification disputes
- Access approval conflicts
- Retention decisions

Level 3

Governance Lead

Issue Types:

- Cross-domain conflicts
- Governance control failures

Level 4

Governance Committee

Issue Types:

- Policy exceptions
- Enterprise governance disputes

Level 5

Executive Governance Sponsor

Issue Types:

- Material governance risk
- Regulatory exposure
- Executive escalation

10. RACI Governance Matrix

Ownership Assignment:

Responsible: Data Steward

Accountable: Data Owner

Consulted: Governance Lead

Informed: Audit Team

Classification Approval:

Responsible: Data Steward

Accountable: Data Owner

Consulted: Security Team

Informed: Governance Office

Access Approval:

Responsible: Access Administrator

Accountable: Data Owner

Consulted: Security Team

Informed: Governance Office

Retention Approval:

Responsible: Data Steward

Accountable: Data Owner

Consulted: Legal/Compliance

Informed: Audit Team

Audit Review:

Responsible: Audit Team

Accountable: Audit Owner

Consulted: Governance Office

Informed: Executive Sponsor

11. Governance Metrics

DG-MET-007

Ownership Assignment Coverage

Target: 100%

DG-MET-008

Steward Assignment Coverage

Target: 100%

DG-MET-009

Ownership Review Completion

Target: $\geq 95\%$

DG-MET-010

Ownership Change Documentation Compliance

Target: 100%

DG-MET-011

Escalation Resolution SLA Compliance

Target: $\geq 95\%$

12. Audit Requirements

AUD-DG-007

Ownership assignments shall be auditable.

AUD-DG-008

Ownership changes shall maintain approval evidence.

AUD-DG-009

Stewardship activities shall be traceable.

AUD-DG-010

Escalation decisions shall be logged.

AUD-DG-011

Governance responsibilities shall be independently verifiable.

13. Compliance Requirements

The ownership framework shall support:

- SOC 2 Accountability Controls
- ISO 27001 Governance Requirements
- ISO 27002 Data Ownership Controls
- Enterprise Procurement Reviews

- Customer Security Assessments
- Internal Audit Programs
- Regulatory Governance Requirements

14. Architectural Constraints

AC-DG-007

Ownership metadata shall be maintained separately from business transaction data.

AC-DG-008

Ownership records shall support historical versioning.

AC-DG-009

Governance assignments shall remain auditable.

AC-DG-010

Ownership repositories shall integrate with Audit and Replay modules.

15. Section Summary

This section establishes the Data Ownership, Stewardship, and Accountability Model for TODP. It defines governance roles, ownership responsibilities, stewardship obligations, escalation mechanisms, accountability structures, domain ownership assignments, governance requirements, audit controls, compliance obligations, and architectural constraints necessary to maintain controlled, traceable, and accountable management of all governed data assets across the platform.

DATA GOVERNANCE V1

Section 3 – Data Classification & Information Categorization Policy

Document ID: GOV-DG-003

Document Name: Data Governance V1

Section ID: DG-03

Section Title: Data Classification & Information Categorization Policy

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Classification and Information Categorization Policy for the Transparent Opportunity Decision Platform (TODP).

The policy defines the classification framework, handling requirements, protection controls, ownership responsibilities, and governance requirements applicable to all data assets managed within TODP.

The objective is to ensure that information is classified according to business value, confidentiality requirements, regulatory obligations, operational criticality, and governance needs throughout the information lifecycle.

2. Objectives

The objectives of this policy are to:

- Establish a consistent enterprise-wide classification model.
- Protect sensitive information from unauthorized disclosure.
- Support regulatory and contractual compliance obligations.
- Define handling requirements for each classification level.
- Enable risk-based security and governance controls.
- Support auditability and traceability requirements.
- Ensure appropriate retention and disposal decisions.
- Improve access governance and least-privilege enforcement.
- Enable secure data sharing across authorized stakeholders.
- Maintain enterprise readiness for audits and customer reviews.

3. Scope

This policy applies to all data assets managed within TODP, including:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- User Information
- Operational Metadata
- System Configuration Data

- Monitoring Data
- Security Event Data
- Backup Data
- Generated Reports
- API Data Exchanges
- Documentation Repositories

The policy applies to:

- Employees
- Contractors
- Third-Party Service Providers
- Platform Administrators
- Governance Teams
- Security Teams
- Audit Teams
- Application Services
- Automated Processing Components

4. Classification Governance Principles

4.1 Classification Before Use

All governed data assets shall be classified before production use.

4.2 Owner Accountability

Data Owners are accountable for classification decisions.

4.3 Least Exposure

Data shall only be exposed to authorized parties based on approved business need.

4.4 Risk-Based Protection

Security controls shall be aligned with classification level.

4.5 Continuous Governance

Classification status shall be periodically reviewed and updated when required.

4.6 Auditability

Classification assignments and modifications shall be auditable.

5. Data Classification Framework

TODP shall use four official classification levels.

Classification Level 1

PUBLIC

Classification Code:

DG-CLASS-PUB

Definition:

Information approved for unrestricted disclosure.

Business Impact if Disclosed:

Negligible

Examples:

- Public marketing materials
- Public product descriptions
- Public documentation
- Approved public communications

Protection Requirements:

- Integrity protection required
- Availability protection required
- No confidentiality restrictions

Approval Authority:

Data Owner

Classification Level 2

INTERNAL

Classification Code:

DG-CLASS-INT

Definition:

Information intended for authorized internal personnel and approved operational use.

Business Impact if Disclosed:

Low

Examples:

- Internal procedures
- Operational dashboards
- Internal reports
- Non-public architecture references

Protection Requirements:

- Access control required
- Integrity monitoring required
- Internal distribution controls

Approval Authority:

Data Owner

Classification Level 3

CONFIDENTIAL

Classification Code:

DG-CLASS-CONF

Definition:

Information whose unauthorized disclosure could result in business, operational, customer, contractual, or compliance impact.

Business Impact if Disclosed:

Moderate to High

Examples:

- Opportunity assessments
- Authority evaluations
- Evidence repositories
- Recommendation records
- Governance decisions

- Internal analytics
- Customer-related information

Protection Requirements:

- Role-based access controls
- Encryption at rest
- Encryption in transit
- Audit logging
- Access reviews

Approval Authority:

Data Owner and Security Owner

Classification Level 4

RESTRICTED

Classification Code:

DG-CLASS-RES

Definition:

Highly sensitive information requiring the strongest governance and protection controls.

Business Impact if Disclosed:

Severe

Examples:

- Security investigations
- Security event records
- Audit evidence repositories
- Sensitive governance investigations
- Cryptographic secrets
- Access control credentials
- Replay integrity records

Protection Requirements:

- Explicit authorization
- Enhanced monitoring
- Multi-layer access controls
- Encryption requirements
- Restricted distribution
- Continuous audit monitoring

Approval Authority:

Data Owner, Security Owner, Governance Office

6. Data Categorization Framework

In addition to classification, data shall be categorized according to business domain.

Category DG-CAT-OPP

Opportunity Data

Category DG-CAT-AUTH

Authority Data

Category DG-CAT-EVD

Evidence Data

Category DG-CAT-RET

Retrieval Data

Category DG-CAT-RANK

Ranking Data

Category DG-CAT-REC

Recommendation Data

Category DG-CAT-GOV

Governance Data

Category DG-CAT-AUD

Audit Data

Category DG-CAT-REP

Replay Data

Category DG-CAT-OPS

Operational Data

Category DG-CAT-SEC

Security Data

Category DG-CAT-CONF

Configuration Data

7. Classification Assignment Process

Step 1

Identify data asset.

Step 2

Identify business purpose.

Step 3

Determine business impact of unauthorized disclosure.

Step 4

Determine integrity requirements.

Step 5

Determine availability requirements.

Step 6

Assign classification level.

Step 7

Assign business category.

Step 8

Record classification metadata.

Step 9

Approve classification.

Step 10

Store classification records in governance repository.

8. Classification Metadata Requirements

The following metadata shall be maintained:

DG-META-001

Data Asset Identifier

DG-META-002

Asset Name

DG-META-003

Data Owner

DG-META-004

Data Steward

DG-META-005

Classification Level

DG-META-006

Business Category

DG-META-007

Creation Date

DG-META-008

Review Date

DG-META-009

Retention Category

DG-META-010

Approval Authority

DG-META-011

Status

9. Reclassification Requirements

DG-CLASS-REQ-001

Classification reviews shall occur annually.

DG-CLASS-REQ-002

Material business changes shall trigger classification review.

DG-CLASS-REQ-003

Regulatory changes shall trigger classification review.

DG-CLASS-REQ-004

Security incidents may trigger immediate reclassification.

DG-CLASS-REQ-005

Reclassification actions shall be documented.

DG-CLASS-REQ-006

Historical classifications shall be retained for audit purposes.

10. Handling Requirements

PUBLIC

Handling Rules:

- May be publicly distributed.
- Integrity controls required.

INTERNAL

Handling Rules:

- Internal access only.
- Controlled sharing required.

CONFIDENTIAL

Handling Rules:

- Approved access only.
- Encryption required.
- Access logging required.
- Sharing approval required.

RESTRICTED

Handling Rules:

- Explicit authorization required.
- Enhanced monitoring required.
- Distribution restrictions required.
- Security review required.

11. Access Governance Requirements

DG-ACCESS-001

Access permissions shall align with classification levels.

DG-ACCESS-002

RESTRICTED data shall require documented approval.

DG-ACCESS-003

CONFIDENTIAL and RESTRICTED data shall maintain audit trails.

DG-ACCESS-004

Periodic access reviews shall validate permissions.

DG-ACCESS-005

Unauthorized access attempts shall be logged and investigated.

12. Governance Rules

DG-GOV-001

All data assets must be classified.

DG-GOV-002

Unclassified production data is prohibited.

DG-GOV-003

Classification metadata shall be maintained throughout the lifecycle.

DG-GOV-004

Classification changes require approval.

DG-GOV-005

Classification records shall be retained for audit purposes.

DG-GOV-006

Classification violations shall be reported to Governance Office.

13. Governance Metrics

DG-MET-012

Classification Coverage

Target: 100%

DG-MET-013

Metadata Completeness

Target: $\geq 98\%$

DG-MET-014

Annual Classification Review Completion

Target: $\geq 95\%$

DG-MET-015

Unauthorized Classification Changes

Target: 0

DG-MET-016

Classification Audit Compliance

Target: $\geq 95\%$

14. Audit Requirements

AUD-DG-012

Classification assignments shall be auditable.

AUD-DG-013

Classification approvals shall maintain evidence records.

AUD-DG-014

Classification changes shall be logged.

AUD-DG-015

Handling controls shall be independently verifiable.

AUD-DG-016

Access permissions shall be traceable to classification requirements.

AUD-DG-017

Historical classification records shall be retained.

15. Compliance Requirements

This policy supports alignment with:

- SOC 2 Security and Confidentiality Criteria
- ISO 27001 Information Classification Controls
- ISO 27002 Data Classification Requirements
- Enterprise Procurement Assessments
- Customer Security Reviews
- Internal Governance Requirements
- Contractual Information Protection Obligations

16. Architectural Constraints

AC-DG-011

Classification metadata shall be stored independently from application business logic.

AC-DG-012

Classification controls shall integrate with access management systems.

AC-DG-013

Classification changes shall generate audit events.

AC-DG-014

Classification repositories shall support historical versioning.

AC-DG-015

Classification services shall integrate with Governance, Audit, and Replay modules.

17. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

18. Risk Considerations

Primary Risks:

- Misclassification
- Unauthorized disclosure
- Regulatory non-compliance
- Excessive access permissions
- Inadequate audit evidence

Mitigation Controls:

- Mandatory classification reviews
- Approval workflows
- Access governance controls
- Audit monitoring
- Governance oversight

19. Control Ownership

Primary Owner:

Governance Office

Supporting Owners:

- Data Owners
- Data Stewards
- Security Team
- Audit Team

Review Frequency:

Annual or upon material change.

20. Section Summary

This section establishes the TODP Data Classification and Information Categorization Policy, defining classification levels, categorization domains, ownership responsibilities, handling requirements, governance controls, audit obligations, compliance requirements, metrics, and architectural constraints necessary to ensure consistent protection, management, and governance of all TODP data assets throughout their lifecycle.

DATA GOVERNANCE V1

Section 4 – Data Lifecycle Management Policy

Document ID: GOV-DG-004

Document Name: Data Governance V1

Section ID: DG-04

Section Title: Data Lifecycle Management Policy

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section defines the Data Lifecycle Management Policy for the Transparent Opportunity Decision Platform (TODP).

The policy establishes governance controls, responsibilities, processes, and requirements governing data throughout its lifecycle, from creation and acquisition through usage, storage, archival, retention, disposal, and destruction.

The objective is to ensure that all TODP data assets remain accurate, traceable, secure, compliant, auditable, and replay-capable throughout their existence.

2. Objectives

The objectives of this policy are to:

- Define a standardized lifecycle model for all governed data assets.
- Ensure consistent governance across all lifecycle stages.
- Support regulatory, contractual, and audit requirements.
- Maintain data quality and integrity throughout processing.
- Protect sensitive information throughout its lifecycle.
- Support replayability and auditability requirements.
- Enable controlled retention and disposal processes.
- Reduce operational and compliance risks.
- Improve accountability for data management activities.
- Support enterprise-scale governance operations.

3. Scope

This policy applies to all TODP data assets including:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- User Activity Data
- Security Event Data
- Configuration Data
- Monitoring Data
- Backup Data
- Reporting Data
- Integration Data

The policy applies to all environments:

- Development
- Testing

- Staging
- Production
- Backup Repositories
- Disaster Recovery Environments
- Archive Storage

4. Lifecycle Governance Principles

4.1 Lifecycle Accountability

Every governed data asset shall have ownership throughout its lifecycle.

4.2 Governance Continuity

Governance controls shall remain active across all lifecycle stages.

4.3 Traceability Preservation

Lineage and auditability shall be maintained throughout the lifecycle.

4.4 Security Preservation

Security protections shall remain aligned with classification requirements.

4.5 Compliance Preservation

Retention and disposal activities shall comply with regulatory obligations.

4.6 Replay Preservation

Replay-critical datasets shall remain available according to replay requirements.

5. Data Lifecycle Model

TODP shall govern data through the following lifecycle stages.

Stage 1

Creation / Acquisition

Lifecycle Code:

DG-LIFE-01

Description:

Data is created, collected, imported, received, generated, or acquired.

Examples:

- Opportunity creation
- Authority ingestion
- Evidence collection
- User submissions
- System-generated records

Required Controls:

- Ownership assignment
- Classification assignment
- Metadata creation
- Validation controls

Stage 2

Validation

Lifecycle Code:

DG-LIFE-02

Description:

Data quality and governance requirements are verified.

Required Controls:

- Data validation
- Schema validation
- Metadata verification
- Quality assessment

Stage 3

Processing

Lifecycle Code:

DG-LIFE-03

Description:

Data is transformed, analyzed, indexed, retrieved, ranked, or used for recommendations.

Required Controls:

- Processing lineage
- Audit logging
- Access controls
- Governance monitoring

Stage 4

Storage

Lifecycle Code:

DG-LIFE-04

Description:

Data is stored in operational repositories.

Required Controls:

- Encryption controls
- Backup controls
- Integrity validation
- Access governance

Stage 5

Operational Use

Lifecycle Code:

DG-LIFE-05

Description:

Data supports platform functionality and business operations.

Required Controls:

- Access monitoring
- Usage logging
- Governance reviews
- Quality monitoring

Stage 6

Archive

Lifecycle Code:

DG-LIFE-06

Description:

Data is moved from active operational use while remaining retained.

Required Controls:

- Archive governance
- Retention enforcement
- Retrieval capability
- Audit preservation

Stage 7

Retention

Lifecycle Code:

DG-LIFE-07

Description:

Data remains preserved according to approved retention schedules.

Required Controls:

- Retention validation
- Legal hold enforcement
- Audit monitoring

Stage 8

Disposal / Destruction

Lifecycle Code:

DG-LIFE-08

Description:

Data reaches end-of-life and is permanently removed according to approved procedures.

Required Controls:

- Destruction approval
- Destruction verification
- Audit evidence
- Governance review

6. Lifecycle Ownership Requirements

DG-LIFE-REQ-001

Every governed dataset shall have an assigned Data Owner.

DG-LIFE-REQ-002

Every governed dataset shall have an assigned Data Steward.

DG-LIFE-REQ-003

Ownership shall remain active throughout all lifecycle stages.

DG-LIFE-REQ-004

Ownership changes shall be auditable.

DG-LIFE-REQ-005

Lifecycle responsibilities shall be documented.

7. Creation and Acquisition Controls

DG-CREATE-001

Data sources shall be identified before ingestion.

DG-CREATE-002

Source authority shall be documented.

DG-CREATE-003

Data classification shall be assigned.

DG-CREATE-004

Metadata records shall be created.

DG-CREATE-005

Lineage tracking shall begin at creation.

DG-CREATE-006

Governance approval shall be required for new critical datasets.

8. Validation Controls

DG-VALID-001

Required fields shall be validated.

DG-VALID-002

Schema compliance shall be verified.

DG-VALID-003

Data quality thresholds shall be enforced.

DG-VALID-004

Classification metadata shall be verified.

DG-VALID-005

Ownership metadata shall be verified.

DG-VALID-006

Validation failures shall be logged.

9. Processing Controls

DG-PROC-001

Processing activities shall be traceable.

DG-PROC-002

Critical transformations shall maintain lineage.

DG-PROC-003

Recommendation generation shall preserve evidence references.

DG-PROC-004

Governance review activities shall be recorded.

DG-PROC-005

Replay reconstruction requirements shall be supported.

DG-PROC-006

Unauthorized processing activities are prohibited.

10. Storage Controls

DG-STOR-001

Data shall be stored according to classification requirements.

DG-STOR-002

Encryption controls shall be applied where required.

DG-STOR-003

Access permissions shall be enforced.

DG-STOR-004

Storage repositories shall support audit logging.

DG-STOR-005

Integrity verification controls shall be implemented.

DG-STOR-006

Storage locations shall be documented.

11. Archive Controls

DG-ARCH-001

Archived data shall remain retrievable.

DG-ARCH-002

Archived records shall preserve metadata.

DG-ARCH-003

Archived audit records shall remain immutable.

DG-ARCH-004

Archived replay records shall remain reproducible.

DG-ARCH-005

Archive repositories shall remain governed.

12. Retention Controls

DG-RET-001

Retention schedules shall be formally approved.

DG-RET-002

Retention periods shall align with governance requirements.

DG-RET-003

Legal holds shall supersede disposal schedules.

DG-RET-004

Retention reviews shall occur periodically.

DG-RET-005

Expired retention candidates shall undergo governance review.

13. Disposal and Destruction Controls

DG-DISP-001

Disposal requires Data Owner approval.

DG-DISP-002

Audit and replay dependencies shall be evaluated before disposal.

DG-DISP-003

Destruction events shall be logged.

DG-DISP-004

Destruction evidence shall be retained.

DG-DISP-005

Unauthorized deletion is prohibited.

DG-DISP-006

Governance approval is required for critical data destruction.

14. Governance Rules

DG-GOV-007

Lifecycle stages shall be documented.

DG-GOV-008

All lifecycle transitions shall be traceable.

DG-GOV-009

Lifecycle controls shall remain auditable.

DG-GOV-010

Critical datasets shall maintain lineage throughout the lifecycle.

DG-GOV-011

Replay-critical datasets shall not be disposed of before approved retention requirements are satisfied.

15. Governance Metrics

DG-MET-017

Lifecycle Metadata Completeness

Target: $\geq 98\%$

DG-MET-018

Ownership Coverage

Target: 100%

DG-MET-019

Retention Compliance

Target: $\geq 95\%$

DG-MET-020

Approved Disposal Compliance

Target: 100%

DG-MET-021

Lineage Preservation Coverage

Target: $\geq 95\%$

DG-MET-022

Archive Retrieval Success Rate

Target: $\geq 99\%$

16. Audit Requirements

AUD-DG-018

Lifecycle transitions shall be auditable.

AUD-DG-019

Retention activities shall maintain evidence.

AUD-DG-020

Destruction approvals shall be documented.

AUD-DG-021

Archive repositories shall be auditable.

AUD-DG-022

Lifecycle ownership assignments shall be verifiable.

AUD-DG-023

Replay dependencies shall be reviewable.

17. Compliance Requirements

This policy supports alignment with:

- SOC 2 Security and Availability Criteria
- SOC 2 Confidentiality Criteria
- ISO 27001 Asset Lifecycle Controls
- ISO 27002 Information Lifecycle Management Controls
- Enterprise Procurement Requirements
- Customer Governance Reviews
- Internal Audit Requirements

18. Architectural Constraints

AC-DG-016

Lifecycle metadata shall be maintained independently from business transaction data.

AC-DG-017

Lifecycle events shall generate audit records.

AC-DG-018

Retention controls shall integrate with Governance, Audit, and Replay modules.

AC-DG-019

Archive repositories shall preserve lineage metadata.

AC-DG-020

Disposal processes shall support verification and evidence generation.

AC-DG-021

Replay-critical datasets shall remain protected from unauthorized destruction.

19. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

20. Section Summary

This section establishes the TODP Data Lifecycle Management Policy, defining lifecycle stages, ownership responsibilities, governance controls, validation requirements, storage requirements, archival processes, retention obligations, disposal controls, audit requirements, compliance obligations, metrics, and architectural constraints necessary to ensure secure, traceable, auditable, and compliant management of data assets throughout their lifecycle.

DATA GOVERNANCE V1

Section 5 – Data Quality Management Framework

Document ID: GOV-DG-005

Document Name: Data Governance V1

Section ID: DG-05

Section Title: Data Quality Management Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Quality Management Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the policies, standards, governance controls, accountability structures, monitoring mechanisms, and quality requirements necessary to ensure that all data assets used within TODP are accurate, complete, consistent, reliable, timely, traceable, and fit for their intended purpose.

High-quality data is essential to ensuring trustworthy Opportunity assessments, Authority evaluations, Evidence integrity, Retrieval effectiveness, Ranking accuracy, Recommendation transparency, Governance oversight, Audit readiness, and Replay reproducibility.

2. Objectives

The objectives of the Data Quality Management Framework are to:

- Establish enterprise-wide data quality standards.
- Define measurable quality dimensions.
- Ensure data reliability across all TODP modules.
- Reduce operational risk caused by poor-quality data.
- Support explainable and defensible recommendations.
- Improve governance decision-making.
- Enable accurate audit and replay operations.
- Establish accountability for data quality outcomes.
- Support regulatory and compliance requirements.
- Maintain enterprise-grade data governance maturity.

3. Scope

This framework applies to:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- Security Data
- Monitoring Data

- Configuration Data
- Integration Data

The framework applies across:

- Data Creation
- Data Acquisition
- Data Processing
- Data Storage
- Data Retrieval
- Data Archival
- Data Reporting
- Data Exchange
- Data Replay Activities

4. Data Quality Governance Principles

4.1 Fitness for Purpose

Data shall be suitable for its intended business use.

4.2 Prevention Over Correction

Quality controls shall prioritize preventing defects rather than correcting them after occurrence.

4.3 Accountability

Data Owners are accountable for quality outcomes.

4.4 Continuous Monitoring

Data quality shall be continuously monitored and measured.

4.5 Traceability

Quality issues shall be traceable to source records and root causes.

4.6 Measurability

Quality standards shall be defined through measurable metrics.

4.7 Auditability

Quality controls and remediation activities shall be auditable.

5. Data Quality Dimensions

TODP shall govern data quality using the following dimensions.

5.1 Accuracy

Quality Code:

DG-DQ-ACC

Definition:

The degree to which data correctly represents the intended real-world value.

Examples:

- Correct opportunity attributes
- Correct authority scores
- Correct evidence references

5.2 Completeness

Quality Code:

DG-DQ-COMP

Definition:

The degree to which required information is present.

Examples:

- Required fields populated
- Mandatory metadata available
- Lineage records complete

5.3 Consistency

Quality Code:

DG-DQ-CONS

Definition:

The degree to which data values remain logically consistent across systems and processes.

Examples:

- Matching identifiers
- Consistent classifications
- Consistent status values

5.4 Timeliness

Quality Code:

DG-DQ-TIME

Definition:

The degree to which data is available when required.

Examples:

- Current opportunity records
- Updated authority sources
- Recent evidence repositories

5.5 Validity

Quality Code:

DG-DQ-VAL

Definition:

The degree to which data conforms to approved formats, structures, and rules.

Examples:

- Schema compliance
- Format validation
- Business rule compliance

5.6 Uniqueness

Quality Code:

DG-DQ-UNIQ

Definition:

The degree to which duplicate records are prevented.

Examples:

- Unique opportunities
- Unique evidence records
- Unique governance records

5.7 Integrity

Quality Code:

DG-DQ-INT

Definition:

The degree to which relationships between data assets remain intact and trustworthy.

Examples:

- Preserved lineage
- Referential integrity
- Traceability continuity

6. Data Quality Roles and Responsibilities

6.1 Data Owner

Responsibilities:

- Quality accountability
- Quality threshold approval

- Remediation prioritization
- Quality governance oversight

6.2 Data Steward

Responsibilities:

- Quality monitoring
- Issue identification
- Remediation coordination
- Quality reporting

6.3 Governance Office

Responsibilities:

- Framework ownership
- Quality standards management
- Governance reporting
- Escalation management

6.4 Engineering Teams

Responsibilities:

- Quality control implementation
- Validation logic maintenance

- Monitoring support

6.5 Audit Team

Responsibilities:

- Quality control verification
- Audit assessments
- Compliance reviews

7. Data Quality Requirements

DG-DQ-REQ-001

All governed datasets shall have defined quality standards.

DG-DQ-REQ-002

Quality thresholds shall be documented.

DG-DQ-REQ-003

Critical datasets shall undergo periodic quality reviews.

DG-DQ-REQ-004

Quality monitoring shall be implemented.

DG-DQ-REQ-005

Quality exceptions shall be documented.

DG-DQ-REQ-006

Root cause analysis shall be performed for material quality incidents.

DG-DQ-REQ-007

Quality remediation actions shall be tracked.

DG-DQ-REQ-008

Quality records shall be auditable.

8. Data Quality Control Framework

Preventive Controls

Purpose:

Prevent quality defects before data enters operational workflows.

Examples:

- Input validation
- Schema enforcement
- Metadata validation
- Classification validation

Detective Controls

Purpose:

Identify quality issues after data creation or processing.

Examples:

- Quality monitoring
- Reconciliation checks
- Exception reporting
- Integrity verification

Corrective Controls

Purpose:

Resolve identified quality issues.

Examples:

- Data correction

- Record remediation
- Metadata repair
- Process improvements

9. Data Quality Monitoring Requirements

DG-MON-001

Quality monitoring shall occur continuously for critical datasets.

DG-MON-002

Quality metrics shall be reviewed periodically.

DG-MON-003

Quality alerts shall be generated when thresholds are violated.

DG-MON-004

Quality dashboards shall be maintained.

DG-MON-005

Material quality issues shall be escalated.

DG-MON-006

Monitoring records shall be retained.

10. Quality Threshold Management

Quality thresholds shall be formally approved by Data Owners.

Threshold categories include:

Critical

Threshold Violation Impact:

High

Required Action:

Immediate remediation.

Major

Threshold Violation Impact:

Moderate

Required Action:

Prioritized remediation.

Minor

Threshold Violation Impact:

Low

Required Action:

Scheduled remediation.

11. Data Quality Issue Management

Issue Lifecycle

Step 1

Issue Identification

Step 2

Issue Classification

Step 3

Impact Assessment

Step 4

Root Cause Analysis

Step 5

Remediation Planning

Step 6

Issue Resolution

Step 7

Verification

Step 8

Closure

All issue lifecycle events shall be recorded.

12. Root Cause Analysis Requirements

DG-RCA-001

Material quality incidents require root cause analysis.

DG-RCA-002

Root causes shall be documented.

DG-RCA-003

Corrective actions shall be identified.

DG-RCA-004

Preventive measures shall be evaluated.

DG-RCA-005

Analysis records shall be retained.

13. Governance Rules

DG-GOV-012

Critical datasets shall maintain approved quality standards.

DG-GOV-013

Quality exceptions shall be documented.

DG-GOV-014

Unresolved critical quality issues shall be escalated.

DG-GOV-015

Quality metrics shall be reported to Governance Office.

DG-GOV-016

Quality controls shall remain auditable.

DG-GOV-017

Quality records shall support replay and audit requirements.

14. Data Quality Metrics

DG-MET-023

Accuracy Compliance

Target: $\geq 98\%$

DG-MET-024

Completeness Compliance

Target: $\geq 98\%$

DG-MET-025

Consistency Compliance

Target: $\geq 97\%$

DG-MET-026

Validity Compliance

Target: $\geq 99\%$

DG-MET-027

Duplicate Record Rate

Target: $\leq 1\%$

DG-MET-028

Quality Issue Resolution SLA

Target: $\geq 95\%$

DG-MET-029

Critical Quality Incident Closure

Target: $\geq 95\%$

15. Audit Requirements

AUD-DG-024

Quality controls shall be auditable.

AUD-DG-025

Quality metrics shall be verifiable.

AUD-DG-026

Issue remediation activities shall maintain evidence.

AUD-DG-027

Root cause analyses shall be retained.

AUD-DG-028

Quality threshold approvals shall be documented.

AUD-DG-029

Monitoring activities shall generate audit records.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Security Criteria
- SOC 2 Processing Integrity Criteria
- ISO 27001 Data Governance Controls
- ISO 27002 Information Quality Controls
- Enterprise Procurement Reviews
- Customer Due Diligence Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-022

Quality monitoring shall not modify source records without approved remediation processes.

AC-DG-023

Quality controls shall preserve lineage integrity.

AC-DG-024

Quality monitoring systems shall integrate with Governance, Audit, and Replay modules.

AC-DG-025

Quality metrics repositories shall support historical analysis.

AC-DG-026

Quality events shall generate auditable records.

AC-DG-027

Quality controls shall support scalability across future data domains.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Inaccurate recommendations
- Incomplete evidence chains
- Governance failures
- Audit deficiencies
- Replay inconsistencies
- Regulatory exposure

Mitigation Controls:

- Continuous monitoring
- Validation controls
- Root cause analysis
- Stewardship reviews
- Governance oversight

20. Section Summary

This section establishes the TODP Data Quality Management Framework, defining quality dimensions, governance responsibilities, monitoring requirements, issue management processes, quality controls, metrics, audit obligations, compliance requirements, and architectural constraints necessary to ensure accurate, complete, reliable, traceable, and enterprise-grade data quality across all TODP modules and governed data assets.

DATA GOVERNANCE V1

Section 6 – Metadata Management & Data Catalog Governance

Document ID: GOV-DG-006

Document Name: Data Governance V1

Section ID: DG-06

Section Title: Metadata Management & Data Catalog Governance

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Metadata Management and Data Catalog Governance Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the standards, governance controls, ownership requirements, catalog structures, metadata lifecycle controls, and operational procedures required to ensure that all governed data assets remain discoverable, understandable, traceable, auditable, and manageable throughout their lifecycle.

Metadata governance serves as a foundational capability supporting transparency, explainability, traceability, auditability, replayability, compliance readiness, and enterprise-scale operations across TODP.

2. Objectives

The objectives of this framework are to:

- Establish standardized metadata governance practices.
- Ensure discoverability of governed data assets.
- Support data ownership and stewardship accountability.
- Enable end-to-end lineage and traceability.
- Improve transparency of data usage and dependencies.
- Support governance, audit, and replay operations.
- Enable enterprise-scale data catalog management.
- Facilitate regulatory and compliance readiness.
- Improve data quality and lifecycle governance.
- Provide a trusted inventory of TODP data assets.

3. Scope

This framework applies to:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- Configuration Data
- Security Data
- Monitoring Data
- Integration Data
- Reporting Data
- Data Storage Assets
- APIs
- Data Pipelines

The framework applies across all environments:

- Development
- Testing
- Staging
- Production
- Archive Repositories

- Backup Repositories
- Disaster Recovery Environments

4. Metadata Governance Principles

4.1 Metadata as a Governed Asset

Metadata shall be governed with the same rigor as business data.

4.2 Catalog-First Governance

All governed data assets shall be registered within approved catalog repositories.

4.3 Ownership Transparency

Metadata shall identify accountable owners and stewards.

4.4 Lifecycle Alignment

Metadata shall remain synchronized with asset lifecycle changes.

4.5 Traceability Preservation

Metadata shall support lineage, audit, and replay requirements.

4.6 Auditability

Metadata changes shall be auditable.

4.7 Standardization

Metadata structures shall follow approved governance standards.

5. Metadata Framework

TODP shall govern three primary metadata domains.

5.1 Business Metadata

Metadata Code:

DG-META-BUS

Purpose:

Describe business meaning and governance context.

Examples:

- Business definitions
- Data owner
- Data steward
- Classification
- Business domain
- Retention category

5.2 Technical Metadata

Metadata Code:

DG-META-TECH

Purpose:

Describe technical implementation details.

Examples:

- Schema definitions
- Data types
- Storage locations
- API mappings
- Pipeline references
- Index structures

5.3 Operational Metadata

Metadata Code:

DG-META-OPS

Purpose:

Describe operational behavior and usage.

Examples:

- Creation timestamps
- Update timestamps
- Processing history
- Access records
- Quality metrics
- Monitoring metrics

6. Data Catalog Governance Model

The TODP Data Catalog shall function as the authoritative inventory of governed data assets.

Catalog Objectives:

- Asset discovery
- Ownership transparency
- Governance visibility
- Lineage support
- Compliance readiness
- Audit support

All critical datasets shall be cataloged before production use.

7. Required Catalog Registration

DG-CAT-001

Opportunity datasets shall be cataloged.

DG-CAT-002

Authority datasets shall be cataloged.

DG-CAT-003

Evidence repositories shall be cataloged.

DG-CAT-004

Retrieval repositories shall be cataloged.

DG-CAT-005

Ranking datasets shall be cataloged.

DG-CAT-006

Recommendation datasets shall be cataloged.

DG-CAT-007

Governance repositories shall be cataloged.

DG-CAT-008

Audit repositories shall be cataloged.

DG-CAT-009

Replay repositories shall be cataloged.

DG-CAT-010

External integration datasets shall be cataloged.

8. Mandatory Metadata Attributes

Every governed asset shall contain the following minimum metadata.

DG-ATTR-001

Asset Identifier

DG-ATTR-002

Asset Name

DG-ATTR-003

Asset Description

DG-ATTR-004

Business Domain

DG-ATTR-005

Data Owner

DG-ATTR-006

Data Steward

DG-ATTR-007

Classification Level

DG-ATTR-008

Retention Category

DG-ATTR-009

Creation Date

DG-ATTR-010

Last Review Date

DG-ATTR-011

Source System

DG-ATTR-012

Lineage Reference

DG-ATTR-013

Status

DG-ATTR-014

Compliance Applicability

9. Metadata Lifecycle Management

Stage 1

Metadata Creation

Requirements:

- Asset registration
- Ownership assignment
- Classification assignment

Stage 2

Metadata Validation

Requirements:

- Completeness validation
- Governance verification

Stage 3

Metadata Maintenance

Requirements:

- Update management
- Change tracking

Stage 4

Metadata Review

Requirements:

- Periodic validation
- Steward review

Stage 5

Metadata Archival

Requirements:

- Historical preservation
- Audit retention

Stage 6

Metadata Disposal

Requirements:

- Approved retirement
- Audit evidence retention

10. Metadata Change Management

DG-CHG-001

Metadata modifications shall be approved.

DG-CHG-002

Ownership changes shall be documented.

DG-CHG-003

Classification changes shall be recorded.

DG-CHG-004

Lineage changes shall be logged.

DG-CHG-005

Catalog updates shall be auditable.

DG-CHG-006

Material changes shall undergo governance review.

11. Metadata Quality Requirements

DG-MQ-001

Metadata completeness shall be monitored.

DG-MQ-002

Metadata accuracy shall be verified.

DG-MQ-003

Metadata consistency shall be maintained.

DG-MQ-004

Metadata duplication shall be minimized.

DG-MQ-005

Metadata reviews shall occur periodically.

DG-MQ-006

Metadata defects shall be remediated.

12. Lineage Metadata Requirements

The catalog shall maintain lineage references for:

- Data Sources
- Data Transformations
- Processing Activities

- Retrieval Operations
- Ranking Operations
- Recommendation Generation
- Governance Reviews
- Audit Activities
- Replay Activities

Lineage metadata shall support end-to-end traceability.

13. Governance Rules

DG-GOV-018

All governed assets shall be cataloged.

DG-GOV-019

Required metadata attributes shall be maintained.

DG-GOV-020

Metadata ownership shall be documented.

DG-GOV-021

Catalog records shall remain auditable.

DG-GOV-022

Metadata shall support traceability requirements.

DG-GOV-023

Metadata repositories shall integrate with governance controls.

DG-GOV-024

Unregistered critical datasets are prohibited in production.

14. Governance Metrics

DG-MET-030

Catalog Coverage

Target: 100%

DG-MET-031

Metadata Completeness

Target: $\geq 98\%$

DG-MET-032

Metadata Accuracy

Target: $\geq 98\%$

DG-MET-033

Ownership Assignment Coverage

Target: 100%

DG-MET-034

Metadata Review Completion

Target: $\geq 95\%$

DG-MET-035

Catalog Registration Compliance

Target: 100%

15. Audit Requirements

AUD-DG-030

Catalog records shall be auditable.

AUD-DG-031

Metadata changes shall maintain history.

AUD-DG-032

Ownership assignments shall be verifiable.

AUD-DG-033

Lineage references shall be reviewable.

AUD-DG-034

Metadata reviews shall maintain evidence.

AUD-DG-035

Catalog completeness shall be independently validated.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Security Criteria

- SOC 2 Processing Integrity Criteria
- ISO 27001 Asset Management Controls
- ISO 27001 Information Governance Controls
- ISO 27002 Metadata Governance Practices
- Enterprise Procurement Reviews
- Customer Governance Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-028

Metadata repositories shall be logically separated from business transaction data.

AC-DG-029

Metadata services shall support version history.

AC-DG-030

Catalog repositories shall support audit logging.

AC-DG-031

Metadata systems shall integrate with Governance, Audit, and Replay modules.

AC-DG-032

Lineage references shall remain immutable after approval.

AC-DG-033

Catalog services shall support future domain expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Undocumented data assets
- Missing ownership assignments
- Incomplete lineage visibility
- Metadata quality degradation
- Audit failures
- Governance gaps

Mitigation Controls:

- Mandatory catalog registration
- Stewardship reviews
- Metadata quality monitoring
- Governance oversight
- Audit verification controls

20. Section Summary

This section establishes the TODP Metadata Management and Data Catalog Governance Framework, defining metadata domains, catalog requirements, ownership responsibilities, lifecycle controls, quality standards, lineage requirements, governance rules, audit obligations, compliance requirements, metrics, and architectural constraints necessary to maintain a trusted, discoverable, traceable, and enterprise-governed inventory of all TODP data assets.

DATA GOVERNANCE V1

Section 7 – Data Lineage, Traceability & Provenance Framework

Document ID: GOV-DG-007

Document Name: Data Governance V1

Section ID: DG-07

Section Title: Data Lineage, Traceability & Provenance Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Lineage, Traceability, and Provenance Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the governance controls, standards, ownership responsibilities, lineage requirements, provenance management practices, and traceability mechanisms necessary to support transparency, explainability, auditability, governance oversight, and replayability across all TODP operations.

Because TODP functions as an explainable decision-support platform, lineage and provenance information are considered critical governance assets required to demonstrate how opportunities are evaluated, how evidence is utilized, how recommendations are generated, and how decisions can be independently verified and reproduced.

2. Objectives

The objectives of this framework are to:

- Establish end-to-end lineage visibility across TODP.
- Support explainable recommendations.
- Enable traceable decision reconstruction.
- Preserve evidence provenance.
- Support governance and audit activities.
- Enable replay and reproducibility capabilities.
- Improve operational transparency.
- Support regulatory and contractual obligations.
- Reduce governance and compliance risk.
- Maintain enterprise-grade traceability controls.

3. Scope

This framework applies to:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records

- Audit Records
- Replay Records
- Metadata Repositories
- Configuration Data
- Security Data
- Integration Data
- Reporting Assets

The framework applies across:

- Data Ingestion
- Data Processing
- Retrieval Operations
- Ranking Operations
- Recommendation Generation
- Governance Reviews
- Audit Activities
- Replay Activities
- Archival Activities

4. Governance Principles

4.1 End-to-End Traceability

All critical data flows shall maintain traceable relationships from source to outcome.

4.2 Provenance Preservation

Original source context shall be preserved throughout processing activities.

4.3 Explainability Support

Lineage records shall support explanation generation.

4.4 Auditability

Lineage and provenance information shall be auditable.

4.5 Replayability

Lineage records shall support reconstruction of historical outcomes.

4.6 Integrity Protection

Lineage records shall be protected against unauthorized modification.

4.7 Accountability

Lineage ownership shall be assigned and governed.

5. Framework Definitions

Data Lineage

Definition:

The recorded path showing how data moves, transforms, and contributes to downstream outcomes.

Traceability

Definition:

The ability to track relationships between data assets, processing activities, governance actions, and decision outcomes.

Provenance

Definition:

The documented origin, history, source authority, and contextual information associated with a data asset.

6. Lineage Coverage Requirements

TODP shall maintain lineage across the following modules.

Module LG-OPP

Opportunity

Required Traceability:

- Source opportunity inputs
- Opportunity transformations
- Opportunity enrichment events

Module LG-AUTH

Authority

Required Traceability:

- Authority source records
- Authority scoring inputs
- Authority validation activities

Module LG-EVD

Evidence

Required Traceability:

- Evidence source origin
- Evidence ingestion records
- Evidence quality assessments

Module LG-RET

Retrieval

Required Traceability:

- Retrieval requests
- Retrieval sources
- Retrieval responses

Module LG-RANK

Ranking

Required Traceability:

- Ranking inputs
- Ranking calculations
- Ranking outputs

Module LG-REC

Recommendation

Required Traceability:

- Recommendation inputs
- Supporting evidence
- Ranking dependencies
- Generated recommendations

Module LG-GOV

Governance

Required Traceability:

- Governance reviews
- Governance approvals
- Governance exceptions

Module LG-AUD

Audit

Required Traceability:

- Audit evidence
- Audit findings
- Audit decisions

Module LG-REP

Replay

Required Traceability:

- Replay datasets
- Replay execution records
- Replay validation outcomes

7. Provenance Management Requirements

DG-PROV-001

All critical datasets shall maintain source attribution.

DG-PROV-002

Evidence records shall preserve source context.

DG-PROV-003

Authority assessments shall maintain source provenance.

DG-PROV-004

Recommendation outputs shall maintain provenance references.

DG-PROV-005

Governance records shall maintain decision provenance.

DG-PROV-006

Replay datasets shall maintain historical provenance.

8. Required Lineage Metadata

The following metadata shall be maintained.

DG-LIN-001

Source Identifier

DG-LIN-002

Source System

DG-LIN-003

Data Asset Identifier

DG-LIN-004

Transformation Identifier

DG-LIN-005

Processing Timestamp

DG-LIN-006

Owner Identifier

DG-LIN-007

Governance Reference

DG-LIN-008

Evidence Reference

DG-LIN-009

Recommendation Reference

DG-LIN-010

Audit Reference

DG-LIN-011

Replay Reference

DG-LIN-012

Lineage Status

9. Lineage Lifecycle Management

Stage 1

Source Registration

Requirements:

- Source identification
- Source approval
- Provenance assignment

Stage 2

Lineage Capture

Requirements:

- Relationship creation
- Metadata generation
- Traceability registration

Stage 3

Lineage Validation

Requirements:

- Completeness review
- Integrity verification

Stage 4

Operational Maintenance

Requirements:

- Change management
- Continuous monitoring

Stage 5

Archival

Requirements:

- Historical preservation
- Audit retention

Stage 6

Retirement

Requirements:

- Governance approval
- Retention validation

10. Traceability Relationships

The following relationships shall be maintained.

Opportunity

→ Authority

Authority

→ Evidence

Evidence

→ Retrieval

Retrieval

→ Ranking

Ranking

→ Recommendation

Recommendation

→ Governance

Governance

→ Audit

Audit

→ Replay

The complete chain shall remain reconstructable.

11. Lineage Integrity Controls

DG-INT-001

Lineage records shall be immutable after approval.

DG-INT-002

Lineage modifications shall be auditable.

DG-INT-003

Integrity verification shall occur periodically.

DG-INT-004

Unauthorized lineage modification is prohibited.

DG-INT-005

Integrity failures shall trigger investigation.

DG-INT-006

Critical lineage repositories shall maintain backup protection.

12. Traceability Requirements

DG-TRACE-001

All critical decisions shall be traceable.

DG-TRACE-002

All recommendations shall reference supporting evidence.

DG-TRACE-003

Governance reviews shall maintain traceability records.

DG-TRACE-004

Audit findings shall reference source evidence.

DG-TRACE-005

Replay activities shall reconstruct traceability chains.

DG-TRACE-006

Cross-domain dependencies shall be documented.

13. Governance Rules

DG-GOV-025

Lineage capture shall be mandatory for critical datasets.

DG-GOV-026

Provenance records shall be retained.

DG-GOV-027

Traceability chains shall support replay requirements.

DG-GOV-028

Lineage repositories shall remain governed.

DG-GOV-029

Lineage exceptions shall be documented.

DG-GOV-030

Lineage integrity shall be monitored.

DG-GOV-031

Traceability controls shall support enterprise audits.

14. Governance Metrics

DG-MET-036

Lineage Coverage

Target: $\geq 95\%$

DG-MET-037

Provenance Completeness

Target: $\geq 98\%$

DG-MET-038

Traceability Verification Success Rate

Target: $\geq 95\%$

DG-MET-039

Lineage Integrity Compliance

Target: $\geq 99\%$

DG-MET-040

Recommendation Traceability Coverage

Target: 100%

DG-MET-041

Replay Traceability Success Rate

Target: $\geq 95\%$

15. Audit Requirements

AUD-DG-036

Lineage records shall be auditable.

AUD-DG-037

Provenance records shall be independently verifiable.

AUD-DG-038

Traceability chains shall support audit reconstruction.

AUD-DG-039

Lineage modifications shall maintain evidence.

AUD-DG-040

Integrity verification results shall be retained.

AUD-DG-041

Replay activities shall demonstrate traceability continuity.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Processing Integrity Criteria
- SOC 2 Security Criteria
- ISO 27001 Governance Controls
- ISO 27001 Asset Traceability Controls

- ISO 27002 Information Management Controls
- Enterprise Procurement Reviews
- Customer Due Diligence Assessments
- Internal Audit Requirements

17. Architectural Constraints

AC-DG-034

Lineage repositories shall be logically separated from transactional processing repositories.

AC-DG-035

Lineage records shall support immutable storage mechanisms.

AC-DG-036

Traceability services shall integrate with Governance, Audit, and Replay modules.

AC-DG-037

Lineage repositories shall support historical versioning.

AC-DG-038

Provenance records shall preserve source attribution throughout the lifecycle.

AC-DG-039

Traceability controls shall not degrade operational system performance beyond approved thresholds.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Broken traceability chains
- Missing provenance records
- Unverifiable recommendations
- Audit reconstruction failures
- Replay inconsistencies
- Governance control failures

Mitigation Controls:

- Mandatory lineage capture
- Provenance governance controls
- Integrity validation
- Audit reviews
- Replay verification procedures
- Governance oversight

20. Section Summary

This section establishes the TODP Data Lineage, Traceability, and Provenance Framework, defining lineage requirements, provenance controls, traceability relationships, ownership responsibilities, governance rules, integrity protections, audit obligations, compliance requirements, metrics, and

architectural constraints necessary to ensure transparent, explainable, auditable, and replayable operation of all TODP decision-support processes and governed data assets.

DATA GOVERNANCE V1

Section 8 – Master Data Management & Reference Data Governance

Document ID: GOV-DG-008

Document Name: Data Governance V1

Section ID: DG-08

Section Title: Master Data Management & Reference Data Governance

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Master Data Management (MDM) and Reference Data Governance Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the governance controls, ownership responsibilities, lifecycle requirements, quality standards, stewardship processes, and operational controls required to ensure that critical master data and reference data assets remain accurate, consistent, authoritative, traceable, and governed across all TODP modules.

The framework supports transparency, explainability, auditability, replayability, and enterprise-scale governance by ensuring that all critical business entities and controlled reference values are managed through approved governance processes.

2. Objectives

The objectives of this framework are to:

- Establish authoritative master data sources.
- Standardize management of reference data assets.
- Improve consistency across TODP modules.
- Reduce duplicate and conflicting records.
- Support traceability and lineage requirements.
- Enable governance and audit readiness.
- Improve data quality and operational reliability.
- Support recommendation explainability.
- Strengthen enterprise data governance maturity.
- Support future scalability and interoperability.

3. Scope

This framework applies to:

Master Data Assets

- Opportunity Entities
- Authority Entities
- Evidence Source Entities
- Governance Entities
- Audit Entities
- Replay Entities
- User Role Entities
- Organizational Entities

- Integration Entities

Reference Data Assets

- Status Codes
- Classification Values
- Governance Categories
- Ranking Categories
- Recommendation Categories
- Retention Categories
- Risk Ratings
- Quality Ratings
- Audit Classifications
- Replay Status Values

The framework applies across:

- Data Creation
- Data Processing
- Data Integration
- Reporting
- Governance Operations
- Audit Activities
- Replay Activities

4. Governance Principles

4.1 Single Source of Truth

Master data shall have one approved authoritative source.

4.2 Controlled Change

Master and reference data modifications shall follow approved governance processes.

4.3 Ownership Accountability

All master and reference data assets shall have assigned ownership.

4.4 Standardization

Reference values shall be standardized and centrally governed.

4.5 Traceability

Changes shall be traceable and auditable.

4.6 Reusability

Master data assets shall be reusable across authorized domains.

4.7 Quality Preservation

Quality controls shall maintain consistency and accuracy.

5. Master Data Framework

Master data represents core business entities shared across TODP.

Master Data Category MDM-OPP

Opportunity Master Data

Examples:

- Opportunity Identifier
- Opportunity Type
- Opportunity Category
- Opportunity Source

Master Data Category MDM-AUTH

Authority Master Data

Examples:

- Authority Identifier
- Authority Classification
- Authority Source Registry

Master Data Category MDM-EVD

Evidence Master Data

Examples:

- Evidence Source Registry
- Evidence Provider Registry
- Evidence Type Definitions

Master Data Category MDM-GOV

Governance Master Data

Examples:

- Governance Control Registry
- Governance Entity Registry
- Governance Committee Registry

Master Data Category MDM-AUD

Audit Master Data

Examples:

- Audit Control Registry
- Audit Classification Registry

Master Data Category MDM-REP

Replay Master Data

Examples:

- Replay Scenario Registry
- Replay Control Registry

6. Reference Data Framework

Reference data consists of controlled values used by business processes.

Reference Data Category REF-CLASS

Classification Values

Examples:

- PUBLIC
- INTERNAL
- CONFIDENTIAL
- RESTRICTED

Reference Data Category REF-STATUS

Status Values

Examples:

- Draft
- Active
- Approved
- Archived
- Retired

Reference Data Category REF-RISK

Risk Ratings

Examples:

- Low
- Medium
- High
- Critical

Reference Data Category REF-RET

Retention Categories

Examples:

- Operational
- Governance
- Audit
- Replay

Reference Data Category REF-QUALITY

Quality Ratings

Examples:

- Excellent
- Acceptable
- Degraded
- Critical

7. Ownership and Stewardship Requirements

DG-MDM-001

All master data assets shall have assigned Data Owners.

DG-MDM-002

All master data assets shall have assigned Data Stewards.

DG-MDM-003

Reference data assets shall have accountable owners.

DG-MDM-004

Ownership records shall be maintained in governance repositories.

DG-MDM-005

Ownership changes shall be approved and documented.

8. Master Data Lifecycle Management

Stage 1

Creation

Requirements:

- Business approval
- Ownership assignment
- Metadata creation

Stage 2

Validation

Requirements:

- Quality verification
- Governance review

Stage 3

Publication

Requirements:

- Approval workflow
- Catalog registration

Stage 4

Operational Use

Requirements:

- Usage monitoring
- Quality controls

Stage 5

Maintenance

Requirements:

- Periodic review
- Change management

Stage 6

Retirement

Requirements:

- Governance approval
- Audit retention validation

9. Reference Data Change Management

DG-REF-001

Reference value creation requires approval.

DG-REF-002

Reference value modification requires governance review.

DG-REF-003

Reference value retirement shall be documented.

DG-REF-004

Historical values shall be retained.

DG-REF-005

Reference changes shall be auditable.

DG-REF-006

Reference repositories shall support version history.

10. Data Quality Requirements

DG-MDM-QA-001

Master data accuracy shall be monitored.

DG-MDM-QA-002

Duplicate master records shall be prevented.

DG-MDM-QA-003

Reference values shall remain standardized.

DG-MDM-QA-004

Master data completeness shall be monitored.

DG-MDM-QA-005

Quality exceptions shall be documented.

DG-MDM-QA-006

Remediation activities shall be tracked.

11. Traceability Requirements

DG-MDM-TRACE-001

Master data changes shall maintain lineage.

DG-MDM-TRACE-002

Reference data changes shall be auditable.

DG-MDM-TRACE-003

Source authority shall be documented.

DG-MDM-TRACE-004

Master data dependencies shall be traceable.

DG-MDM-TRACE-005

Replay activities shall preserve applicable master data references.

DG-MDM-TRACE-006

Audit reviews shall support historical reconstruction.

12. Governance Rules

DG-GOV-032

Authoritative sources shall be documented.

DG-GOV-033

Duplicate master records are prohibited.

DG-GOV-034

Reference values shall be centrally governed.

DG-GOV-035

Master data repositories shall remain cataloged.

DG-GOV-036

Master data modifications shall follow approved workflows.

DG-GOV-037

Reference data modifications shall be controlled.

DG-GOV-038

Governance exceptions shall be documented.

13. Governance Metrics

DG-MET-042

Master Data Ownership Coverage

Target: 100%

DG-MET-043

Reference Data Standardization Compliance

Target: $\geq 99\%$

DG-MET-044

Duplicate Master Record Rate

Target: $\leq 1\%$

DG-MET-045

Master Data Quality Compliance

Target: $\geq 98\%$

DG-MET-046

Reference Change Approval Compliance

Target: 100%

DG-MET-047

Master Data Review Completion

Target: $\geq 95\%$

14. Audit Requirements

AUD-DG-042

Master data ownership shall be verifiable.

AUD-DG-043

Reference data modifications shall be auditable.

AUD-DG-044

Master data lineage shall support reconstruction.

AUD-DG-045

Authoritative source assignments shall be documented.

AUD-DG-046

Quality reviews shall maintain evidence.

AUD-DG-047

Historical versions shall remain accessible.

15. Compliance Requirements

This framework supports alignment with:

- SOC 2 Security Criteria
- SOC 2 Processing Integrity Criteria
- ISO 27001 Governance Controls
- ISO 27001 Asset Management Controls
- ISO 27002 Data Management Controls
- Enterprise Procurement Assessments
- Customer Governance Reviews
- Internal Audit Programs

16. Architectural Constraints

AC-DG-040

Master data repositories shall be logically separated from transactional repositories.

AC-DG-041

Reference data repositories shall support version control.

AC-DG-042

Master data services shall integrate with Metadata, Governance, Audit, and Replay modules.

AC-DG-043

Master data identifiers shall remain stable throughout the lifecycle.

AC-DG-044

Historical reference values shall be retained.

AC-DG-045

Authoritative source assignments shall support future platform expansion.

17. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

18. Risk Considerations

Primary Risks:

- Duplicate master records
- Conflicting reference values
- Inconsistent entity definitions
- Governance control failures
- Audit reconstruction failures
- Replay inconsistencies

Mitigation Controls:

- Single source governance
- Stewardship reviews
- Change approval workflows
- Quality monitoring
- Lineage controls
- Audit verification

19. Control Ownership

Primary Owner:

Governance Office

Supporting Owners:

- Data Owners
- Data Stewards
- Security Team
- Audit Team

Review Frequency:

Annual or upon material governance change.

20. Section Summary

This section establishes the TODP Master Data Management and Reference Data Governance Framework, defining authoritative data sources, stewardship responsibilities, lifecycle controls, change management processes, quality requirements, traceability obligations, governance rules, audit requirements, compliance obligations, metrics, and architectural constraints necessary to ensure consistent, accurate, governed, and reusable master and reference data assets across all TODP modules.

DATA GOVERNANCE V1

Section 9 – Data Retention, Archival & Disposal Governance Policy

Document ID: GOV-DG-009

Document Name: Data Governance V1

Section ID: DG-09

Section Title: Data Retention, Archival & Disposal Governance Policy

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Retention, Archival, and Disposal Governance Policy for the Transparent Opportunity Decision Platform (TODP).

The policy defines the governance framework, retention schedules, archival controls, disposal procedures, ownership responsibilities, audit requirements, and compliance obligations governing the preservation and destruction of TODP data assets.

The objective is to ensure that data is retained for appropriate business, governance, audit, legal, compliance, operational, and replay purposes while minimizing risk associated with excessive retention and unauthorized disposal.

2. Objectives

The objectives of this policy are to:

- Establish standardized retention schedules.
- Define archival governance controls.
- Ensure controlled and auditable data disposal.
- Support governance, audit, and replay requirements.
- Preserve critical historical records.
- Support legal and regulatory obligations.
- Reduce storage and compliance risks.
- Ensure accountability for retention decisions.
- Improve lifecycle governance maturity.
- Maintain enterprise readiness for audits and procurement reviews.

3. Scope

This policy applies to:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- Security Event Data
- Monitoring Data

- User Activity Data
- Configuration Data
- Backup Data
- Archive Data
- Reporting Data

The policy applies across:

- Production Systems
- Archive Repositories
- Backup Repositories
- Disaster Recovery Environments
- Governance Repositories
- Audit Repositories
- Replay Repositories

4. Governance Principles

4.1 Retain for Business Need

Data shall be retained only for approved business, governance, audit, legal, operational, or replay purposes.

4.2 Controlled Disposal

Data disposal shall occur only through approved governance processes.

4.3 Auditability

Retention and disposal activities shall be auditable.

4.4 Replay Preservation

Replay-critical records shall remain available according to replay requirements.

4.5 Legal Compliance

Legal and regulatory obligations shall supersede standard retention schedules.

4.6 Ownership Accountability

Data Owners are accountable for retention decisions.

4.7 Traceability

Retention and disposal activities shall maintain traceable evidence.

5. Retention Governance Framework

TODP shall classify retention requirements into the following categories.

Category DG-RET-OPR

Operational Retention

Purpose:

Support ongoing business and platform operations.

Category DG-RET-GOV

Governance Retention

Purpose:

Support governance oversight and policy compliance.

Category DG-RET-AUD

Audit Retention

Purpose:

Support internal and external audit requirements.

Category DG-RET-REP

Replay Retention

Purpose:

Support decision reconstruction and replay operations.

Category DG-RET-SEC

Security Retention

Purpose:

Support incident investigation and security monitoring.

Category DG-RET-LEGAL

Legal Retention

Purpose:

Support legal and regulatory obligations.

6. Standard Retention Schedule

Opportunity Data

Retention Category:

Operational

Minimum Retention:

5 Years

Authority Data

Retention Category:

Operational

Minimum Retention:

5 Years

Evidence Data

Retention Category:

Operational / Replay

Minimum Retention:

7 Years

Retrieval Records

Retention Category:

Operational

Minimum Retention:

3 Years

Ranking Records

Retention Category:

Replay

Minimum Retention:

7 Years

Recommendation Records

Retention Category:

Replay

Minimum Retention:

7 Years

Governance Records

Retention Category:

Governance

Minimum Retention:

7 Years

Audit Records

Retention Category:

Audit

Minimum Retention:

10 Years

Replay Records

Retention Category:

Replay

Minimum Retention:

10 Years

Metadata Records

Retention Category:

Governance

Minimum Retention:

7 Years

Security Event Records

Retention Category:

Security

Minimum Retention:

5 Years

Configuration History

Retention Category:

Governance

Minimum Retention:

5 Years

7. Legal Hold Requirements

DG-HOLD-001

Legal holds shall override disposal schedules.

DG-HOLD-002

Legal hold records shall be documented.

DG-HOLD-003

Affected data assets shall be protected from destruction.

DG-HOLD-004

Legal hold releases require documented authorization.

DG-HOLD-005

Legal hold actions shall be auditable.

DG-HOLD-006

Governance Office shall be notified of active legal holds.

8. Archival Governance Framework

Archived data shall:

- Remain retrievable.
- Preserve metadata.
- Preserve lineage references.
- Preserve classification information.
- Maintain auditability.
- Support replay reconstruction where required.

Archive repositories shall remain governed assets.

9. Archive Requirements

DG-ARCH-006

Archived records shall retain ownership metadata.

DG-ARCH-007

Archived records shall retain classification metadata.

DG-ARCH-008

Archived audit records shall remain immutable.

DG-ARCH-009

Archived replay records shall remain reproducible.

DG-ARCH-010

Archive access shall be controlled.

DG-ARCH-011

Archive integrity shall be monitored.

10. Disposal Governance Framework

Disposal shall occur only after:

- Retention period expiration.
- Legal hold validation.
- Governance review.
- Ownership approval.
- Dependency assessment.

Disposal activities shall generate audit evidence.

11. Disposal Approval Requirements

DG-DISP-007

Data Owner approval is required.

DG-DISP-008

Governance review is required for critical datasets.

DG-DISP-009

Audit dependencies shall be evaluated.

DG-DISP-010

Replay dependencies shall be evaluated.

DG-DISP-011

Disposal requests shall be documented.

DG-DISP-012

Approvals shall be retained.

12. Secure Destruction Requirements

DG-DEST-001

Disposed data shall be permanently removed using approved methods.

DG-DEST-002

Destruction verification shall be performed.

DG-DEST-003

Destruction evidence shall be retained.

DG-DEST-004

Unauthorized destruction is prohibited.

DG-DEST-005

Critical record destruction shall require enhanced approval.

DG-DEST-006

Destruction activities shall be auditable.

13. Governance Rules

DG-GOV-039

Retention schedules shall be documented.

DG-GOV-040

Retention schedules shall be periodically reviewed.

DG-GOV-041

Retention exceptions shall be approved.

DG-GOV-042

Archived records shall remain governed.

DG-GOV-043

Disposal activities shall maintain evidence.

DG-GOV-044

Replay-critical records shall not be prematurely destroyed.

DG-GOV-045

Legal hold requirements shall supersede disposal schedules.

14. Governance Metrics

DG-MET-048

Retention Compliance

Target: $\geq 95\%$

DG-MET-049

Archive Retrieval Success Rate

Target: $\geq 99\%$

DG-MET-050

Approved Disposal Compliance

Target: 100%

DG-MET-051

Legal Hold Compliance

Target: 100%

DG-MET-052

Archive Integrity Compliance

Target: $\geq 99\%$

DG-MET-053

Destruction Evidence Completion

Target: 100%

15. Audit Requirements

AUD-DG-048

Retention schedules shall be auditable.

AUD-DG-049

Archive repositories shall be reviewable.

AUD-DG-050

Disposal approvals shall maintain evidence.

AUD-DG-051

Legal hold records shall be verifiable.

AUD-DG-052

Destruction activities shall maintain audit records.

AUD-DG-053

Replay retention obligations shall be independently validated.

16. Compliance Requirements

This policy supports alignment with:

- SOC 2 Availability Criteria
- SOC 2 Security Criteria
- SOC 2 Confidentiality Criteria
- ISO 27001 Information Retention Controls
- ISO 27001 Asset Management Controls
- ISO 27002 Information Disposal Controls
- Enterprise Procurement Reviews
- Customer Governance Assessments

17. Architectural Constraints

AC-DG-046

Archive repositories shall be logically separated from production repositories.

AC-DG-047

Retention services shall integrate with Governance, Audit, and Replay modules.

AC-DG-048

Archive repositories shall preserve metadata and lineage references.

AC-DG-049

Disposal workflows shall generate immutable audit records.

AC-DG-050

Retention controls shall support historical reconstruction requirements.

AC-DG-051

Archive systems shall support future retention schedule expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Premature data deletion
- Excessive data retention
- Legal non-compliance
- Archive corruption
- Replay reconstruction failures
- Audit evidence loss

Mitigation Controls:

- Approved retention schedules
- Legal hold controls
- Archive integrity monitoring
- Disposal approval workflows
- Audit verification
- Governance oversight

20. Section Summary

This section establishes the TODP Data Retention, Archival, and Disposal Governance Policy, defining retention schedules, archival requirements, legal hold controls, disposal procedures, destruction standards, governance responsibilities, audit obligations, compliance requirements, metrics, and architectural constraints necessary to ensure secure, compliant, auditable, and replay-capable management of data assets throughout their retention lifecycle.

DATA GOVERNANCE V1

Section 10 – Data Access Governance & Authorization Management

Document ID: GOV-DG-010

Document Name: Data Governance V1

Section ID: DG-10

Section Title: Data Access Governance & Authorization Management

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Access Governance and Authorization Management Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines governance controls, access approval processes, authorization models, accountability requirements, monitoring obligations, audit controls, and compliance requirements governing access to TODP data assets.

The objective is to ensure that access to data is granted, reviewed, monitored, modified, and revoked in a controlled, auditable, risk-based, and least-privilege manner while preserving transparency, explainability, governance integrity, auditability, and replayability.

2. Objectives

The objectives of this framework are to:

- Establish controlled access governance practices.
- Enforce least-privilege access principles.
- Protect sensitive and regulated information.
- Ensure accountability for access decisions.
- Support governance and audit readiness.
- Reduce unauthorized access risk.
- Strengthen operational security controls.
- Support compliance obligations.
- Improve access transparency and traceability.
- Maintain enterprise-grade authorization governance.

3. Scope

This framework applies to access involving:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- Security Data
- Configuration Data

- Monitoring Data
- Archive Data
- Backup Data

The framework applies to:

- Employees
- Contractors
- Administrators
- Governance Personnel
- Audit Personnel
- Service Accounts
- System Components
- Third-Party Integrations

4. Governance Principles

4.1 Least Privilege

Access shall be limited to the minimum permissions required to perform authorized responsibilities.

4.2 Need-to-Know

Access shall be granted only when supported by legitimate business requirements.

4.3 Segregation of Duties

Conflicting privileges shall be restricted through governance controls.

4.4 Ownership Accountability

Data Owners are accountable for access authorization decisions.

4.5 Continuous Review

Access permissions shall be periodically reviewed.

4.6 Traceability

Access activities shall be traceable and auditable.

4.7 Default Deny

Access shall be denied unless explicitly authorized.

5. Access Governance Model

TODP shall implement a governance-based authorization model consisting of:

Layer 1

Identity Verification

Purpose:

Validate user or system identity.

Layer 2

Role Authorization

Purpose:

Assign permissions according to approved roles.

Layer 3

Data Authorization

Purpose:

Control access to governed datasets.

Layer 4

Governance Enforcement

Purpose:

Apply ownership, classification, and policy requirements.

Layer 5

Audit Monitoring

Purpose:

Monitor and verify access activities.

6. Access Role Categories

Role Category DG-ROLE-USER

Business User

Typical Access:

- Approved operational data
- Assigned responsibilities

Role Category DG-ROLE-ANALYST

Analyst

Typical Access:

- Retrieval outputs
- Evidence repositories

- Recommendation outputs

Role Category DG-ROLE-GOV

Governance Personnel

Typical Access:

- Governance repositories
- Policy records
- Review workflows

Role Category DG-ROLE-AUD

Audit Personnel

Typical Access:

- Audit repositories
- Audit evidence
- Compliance records

Role Category DG-ROLE-SEC

Security Personnel

Typical Access:

- Security repositories
- Monitoring systems
- Investigation records

Role Category DG-ROLE-ADMIN

Platform Administrator

Typical Access:

- Platform administration
- Infrastructure management

Role Category DG-ROLE-SVC

Service Account

Typical Access:

- Approved automated processing activities

7. Access Approval Requirements

DG-ACCESS-006

All access requests shall be documented.

DG-ACCESS-007

Business justification shall be provided.

DG-ACCESS-008

Data Owner approval shall be required.

DG-ACCESS-009

Restricted access requires additional review.

DG-ACCESS-010

Approvals shall be retained.

DG-ACCESS-011

Unauthorized access requests shall be denied.

8. Classification-Based Access Controls

PUBLIC

Access Requirement:

Approved operational access.

INTERNAL

Access Requirement:

Authenticated internal access.

CONFIDENTIAL

Access Requirement:

Data Owner approval.

Role validation.

Audit logging.

RESTRICTED

Access Requirement:

Data Owner approval.

Security approval.

Enhanced monitoring.

Audit logging.

Periodic review.

9. Segregation of Duties Requirements

DG-SOD-001

Access request approval and access provisioning shall be separated.

DG-SOD-002

Governance review and audit review responsibilities shall remain independent.

DG-SOD-003

Security monitoring and access approval responsibilities shall remain independent.

DG-SOD-004

No individual shall approve their own access request.

DG-SOD-005

Conflicting privileges shall be identified and controlled.

DG-SOD-006

Segregation conflicts shall be documented and reviewed.

10. Access Review Requirements

DG-REV-001

Access reviews shall occur quarterly for critical datasets.

DG-REV-002

Restricted access permissions shall be reviewed quarterly.

DG-REV-003

Governance access permissions shall be reviewed semi-annually.

DG-REV-004

Audit access permissions shall be reviewed semi-annually.

DG-REV-005

Inactive accounts shall be evaluated.

DG-REV-006

Review evidence shall be retained.

11. Access Modification and Revocation

DG-MOD-001

Role changes shall trigger access review.

DG-MOD-002

Termination events shall trigger access revocation.

DG-MOD-003

Contract expiration shall trigger access review.

DG-MOD-004

Privilege escalation requires approval.

DG-MOD-005

Revocation activities shall be logged.

DG-MOD-006

Revocation records shall be auditable.

12. Monitoring Requirements

DG-MON-007

Access activities shall be logged.

DG-MON-008

Unauthorized access attempts shall be monitored.

DG-MON-009

Privileged activities shall be monitored.

DG-MON-010

Restricted data access shall be monitored.

DG-MON-011

Anomalous access behavior shall be investigated.

DG-MON-012

Monitoring records shall be retained.

13. Governance Rules

DG-GOV-046

Access shall follow least-privilege principles.

DG-GOV-047

Access approvals shall be documented.

DG-GOV-048

Periodic reviews shall be performed.

DG-GOV-049

Privileged access shall be monitored.

DG-GOV-050

Unauthorized access shall be investigated.

DG-GOV-051

Restricted data access shall require enhanced governance controls.

DG-GOV-052

Access records shall support audit and replay requirements.

14. Governance Metrics

DG-MET-054

Access Review Completion

Target: $\geq 95\%$

DG-MET-055

Unauthorized Access Incidents

Target: 0

DG-MET-056

Access Approval Documentation Compliance

Target: 100%

DG-MET-057

Privileged Access Review Compliance

Target: $\geq 95\%$

DG-MET-058

Revocation SLA Compliance

Target: $\geq 95\%$

DG-MET-059

Segregation of Duties Compliance

Target: $\geq 95\%$

15. Audit Requirements

AUD-DG-054

Access approvals shall be auditable.

AUD-DG-055

Access reviews shall maintain evidence.

AUD-DG-056

Privileged access activities shall be reviewable.

AUD-DG-057

Revocation activities shall be traceable.

AUD-DG-058

Segregation of duties controls shall be verifiable.

AUD-DG-059

Access governance metrics shall be independently validated.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Security Criteria
- SOC 2 Confidentiality Criteria
- ISO 27001 Access Control Requirements
- ISO 27001 Identity Governance Controls
- ISO 27002 Authorization Management Controls
- Enterprise Procurement Assessments
- Customer Security Reviews
- Internal Audit Programs

17. Architectural Constraints

AC-DG-052

Authorization controls shall integrate with identity management services.

AC-DG-053

Access governance records shall be stored separately from business transaction data.

AC-DG-054

Access activities shall generate immutable audit records.

AC-DG-055

Access governance services shall integrate with Governance, Audit, and Replay modules.

AC-DG-056

Role assignments shall support version history.

AC-DG-057

Authorization controls shall support future organizational growth.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Unauthorized access
- Excessive privileges
- Segregation of duties violations
- Data disclosure incidents
- Governance failures
- Audit findings

Mitigation Controls:

- Approval workflows
- Periodic reviews
- Privileged access monitoring
- Role governance
- Security oversight
- Audit verification

20. Section Summary

This section establishes the TODP Data Access Governance and Authorization Management Framework, defining authorization models, role governance, approval requirements, segregation of duties controls, review processes, monitoring obligations, governance rules, audit requirements, compliance obligations, metrics, and architectural constraints necessary to ensure controlled, traceable, auditable, and least-privilege access to all governed TODP data assets.

DATA GOVERNANCE V1

Section 11 – Data Privacy Governance & Sensitive Information Management

Document ID: GOV-DG-011

Document Name: Data Governance V1

Section ID: DG-11

Section Title: Data Privacy Governance & Sensitive Information Management

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Privacy Governance and Sensitive Information Management Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the governance controls, privacy principles, ownership responsibilities, handling requirements, protection measures, audit obligations, and compliance controls required to manage sensitive information throughout its lifecycle.

The objective is to ensure that TODP processes, stores, transmits, archives, and disposes of sensitive information in a lawful, controlled, transparent, auditable, and risk-based manner while supporting enterprise governance, audit readiness, replayability, and future regulatory compliance obligations.

2. Objectives

The objectives of this framework are to:

- Establish privacy governance standards.
- Protect sensitive information from unauthorized disclosure.
- Support lawful and compliant data processing.
- Minimize privacy-related risks.
- Define sensitive information handling requirements.
- Strengthen governance and accountability.
- Support audit and compliance reviews.
- Improve transparency regarding data usage.
- Enable controlled access and processing.
- Maintain enterprise-grade privacy governance maturity.

3. Scope

This framework applies to:

- User Information
- Identity Information
- Authentication Data
- Authorization Data
- Opportunity Records
- Authority Records
- Evidence Records
- Governance Records
- Audit Records
- Replay Records
- Security Event Records
- Monitoring Records

- Metadata Repositories
- Integration Data
- Reporting Data

The framework applies across:

- Collection Activities
- Processing Activities
- Storage Activities
- Transmission Activities
- Reporting Activities
- Archival Activities
- Disposal Activities

4. Privacy Governance Principles

4.1 Lawful Processing

Information shall be processed only for approved and legitimate purposes.

4.2 Purpose Limitation

Information shall be used only for approved business objectives.

4.3 Data Minimization

Only the minimum necessary information shall be collected and processed.

4.4 Accuracy

Sensitive information shall remain accurate and current where applicable.

4.5 Storage Limitation

Sensitive information shall not be retained beyond approved retention requirements.

4.6 Security Protection

Sensitive information shall be protected according to classification and risk requirements.

4.7 Accountability

Privacy governance activities shall have assigned accountability.

5. Sensitive Information Categories

Category DG-PRIV-001

Identity Information

Examples:

- User identifiers
- Account identifiers

- Internal personnel identifiers

Category DG-PRIV-002

Authentication Information

Examples:

- Authentication records
- Credential references
- Identity verification records

Category DG-PRIV-003

Authorization Information

Examples:

- Access assignments
- Role mappings
- Permission records

Category DG-PRIV-004

Operationally Sensitive Information

Examples:

- Governance investigations
- Security investigations
- Audit evidence

Category DG-PRIV-005

Restricted Security Information

Examples:

- Security event records
- Security control records
- Threat investigation data

6. Privacy Governance Roles

Executive Governance Sponsor

Responsibilities:

- Privacy governance oversight
- Risk acceptance authority

Governance Office

Responsibilities:

- Privacy framework ownership
- Governance reporting

Data Owners

Responsibilities:

- Privacy classification approval
- Processing authorization

Data Stewards

Responsibilities:

- Privacy control execution
- Metadata maintenance

Security Team

Responsibilities:

- Protection controls
- Security monitoring

Audit Team

Responsibilities:

- Independent verification
- Privacy control assessments

7. Sensitive Information Handling Requirements

DG-PRIV-HND-001

Sensitive information shall be classified before operational use.

DG-PRIV-HND-002

Access shall follow least-privilege principles.

DG-PRIV-HND-003

Sensitive information shall be encrypted where required.

DG-PRIV-HND-004

Handling activities shall be auditable.

DG-PRIV-HND-005

Unauthorized disclosure is prohibited.

DG-PRIV-HND-006

Sensitive information repositories shall remain governed.

8. Data Collection Governance

DG-PRIV-COL-001

Collection purposes shall be documented.

DG-PRIV-COL-002

Collected information shall support approved business requirements.

DG-PRIV-COL-003

Collection shall follow data minimization principles.

DG-PRIV-COL-004

Ownership shall be assigned at collection.

DG-PRIV-COL-005

Classification shall be assigned during onboarding.

DG-PRIV-COL-006

Collection activities shall be traceable.

9. Data Usage Governance

DG-PRIV-USE-001

Information shall only be used for approved purposes.

DG-PRIV-USE-002

Secondary usage requires governance review.

DG-PRIV-USE-003

Access shall be authorized.

DG-PRIV-USE-004

Usage activities shall be logged.

DG-PRIV-USE-005

Unauthorized sharing is prohibited.

DG-PRIV-USE-006

Usage records shall support audits.

10. Data Sharing Governance

DG-PRIV-SHR-001

Sharing shall require documented authorization.

DG-PRIV-SHR-002

Classification requirements shall be evaluated before sharing.

DG-PRIV-SHR-003

Sharing recipients shall be approved.

DG-PRIV-SHR-004

Sharing activities shall be logged.

DG-PRIV-SHR-005

Restricted information sharing requires enhanced approval.

DG-PRIV-SHR-006

Sharing records shall be retained.

11. Data Subject and Information Requests

The platform shall support governance procedures for:

- Information access requests
- Information correction requests
- Information deletion requests
- Governance inquiries
- Audit requests
- Compliance reviews

All requests shall be documented and tracked.

12. Privacy Risk Management

DG-PRIV-RISK-001

Privacy risks shall be periodically assessed.

DG-PRIV-RISK-002

Material privacy risks shall be documented.

DG-PRIV-RISK-003

Risk mitigation plans shall be maintained.

DG-PRIV-RISK-004

Risk reviews shall occur periodically.

DG-PRIV-RISK-005

Privacy incidents shall be escalated.

DG-PRIV-RISK-006

Risk records shall be retained.

13. Governance Rules

DG-GOV-053

Sensitive information shall be classified.

DG-GOV-054

Unauthorized collection is prohibited.

DG-GOV-055

Unauthorized processing is prohibited.

DG-GOV-056

Sensitive information sharing shall be controlled.

DG-GOV-057

Privacy governance activities shall be auditable.

DG-GOV-058

Privacy exceptions shall be documented.

DG-GOV-059

Privacy controls shall support audit and replay requirements.

14. Governance Metrics

DG-MET-060

Sensitive Information Classification Coverage

Target: 100%

DG-MET-061

Access Review Compliance

Target: $\geq 95\%$

DG-MET-062

Privacy Incident Resolution Compliance

Target: $\geq 95\%$

DG-MET-063

Sensitive Data Encryption Compliance

Target: 100%

DG-MET-064

Approved Sharing Compliance

Target: 100%

DG-MET-065

Privacy Risk Review Completion

Target: $\geq 95\%$

15. Audit Requirements

AUD-DG-060

Sensitive information inventories shall be auditable.

AUD-DG-061

Privacy approvals shall maintain evidence.

AUD-DG-062

Sharing activities shall be reviewable.

AUD-DG-063

Privacy incidents shall maintain investigation records.

AUD-DG-064

Risk assessments shall be retained.

AUD-DG-065

Privacy governance controls shall be independently verified.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Confidentiality Criteria
- SOC 2 Privacy Criteria
- ISO 27001 Information Protection Controls
- ISO 27001 Privacy Governance Controls
- ISO 27002 Sensitive Information Controls
- Enterprise Procurement Assessments
- Customer Security Reviews
- Internal Audit Programs

Future regulatory alignment may include:

- GDPR-related obligations
- Privacy-by-design principles
- Customer contractual privacy requirements

17. Architectural Constraints

AC-DG-058

Sensitive information repositories shall support classification-aware controls.

AC-DG-059

Privacy governance controls shall integrate with Access Governance services.

AC-DG-060

Privacy events shall generate audit records.

AC-DG-061

Sensitive information lineage shall remain traceable.

AC-DG-062

Privacy governance services shall integrate with Governance, Audit, and Replay modules.

AC-DG-063

Sensitive information controls shall support future regulatory expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Unauthorized disclosure
- Privacy violations
- Excessive data collection
- Improper sharing
- Governance failures
- Compliance findings

Mitigation Controls:

- Classification controls
- Access governance
- Encryption controls
- Sharing approvals
- Privacy risk reviews
- Audit verification

20. Section Summary

This section establishes the TODP Data Privacy Governance and Sensitive Information Management Framework, defining privacy governance principles, sensitive information categories, collection and usage controls, sharing requirements, risk management processes, governance rules, audit obligations, compliance requirements, metrics, and architectural constraints necessary to ensure lawful, controlled, auditable, and enterprise-grade management of sensitive information across all TODP modules and governed data assets.

DATA GOVERNANCE V1

Section 12 – Data Security Governance & Protection Controls Framework

Document ID: GOV-DG-012

Document Name: Data Governance V1

Section ID: DG-12

Section Title: Data Security Governance & Protection Controls Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Security Governance and Protection Controls Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the governance model, security control requirements, accountability structures, protection mechanisms, monitoring obligations, audit requirements, and compliance controls necessary to safeguard TODP data assets against unauthorized access, disclosure, modification, destruction, corruption, or misuse.

The objective is to ensure the confidentiality, integrity, availability, traceability, auditability, and replayability of all governed data assets while supporting enterprise readiness, customer trust, operational resilience, and future security certification objectives.

2. Objectives

The objectives of this framework are to:

- Establish enterprise-grade data security governance.
- Protect data assets throughout their lifecycle.
- Enforce classification-based security controls.
- Support secure operation of TODP modules.
- Reduce data-related security risks.
- Strengthen governance accountability.
- Enable security monitoring and response.
- Support audit and compliance requirements.
- Improve operational resilience.
- Maintain alignment with enterprise security expectations.

3. Scope

This framework applies to:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- Security Event Data
- User Activity Data
- Configuration Data
- Archive Data
- Backup Data
- Integration Data

The framework applies across:

- Data Collection
- Data Processing
- Data Storage
- Data Transmission
- Data Sharing

- Data Archival
- Data Disposal
- Recovery Operations

4. Security Governance Principles

4.1 Confidentiality

Data shall be protected against unauthorized disclosure.

4.2 Integrity

Data shall be protected against unauthorized modification.

4.3 Availability

Data shall remain available according to business requirements.

4.4 Defense in Depth

Multiple layers of protection shall be implemented.

4.5 Least Privilege

Access shall be restricted to authorized users and systems.

4.6 Continuous Monitoring

Security events shall be continuously monitored.

4.7 Accountability

Security responsibilities shall be formally assigned.

5. Security Governance Structure

Executive Governance Sponsor

Responsibilities:

- Security oversight
- Risk acceptance decisions

Governance Office

Responsibilities:

- Security governance coordination
- Policy oversight

Security Team

Responsibilities:

- Security control implementation
- Monitoring and response

Data Owners

Responsibilities:

- Classification approval
- Access authorization approval

Data Stewards

Responsibilities:

- Security governance execution
- Control validation

Audit Team

Responsibilities:

- Independent control verification

6. Data Protection Control Domains

Control Domain DG-SEC-001

Access Protection

Purpose:

Protect data from unauthorized access.

Control Domain DG-SEC-002

Encryption Protection

Purpose:

Protect data confidentiality.

Control Domain DG-SEC-003

Integrity Protection

Purpose:

Protect data accuracy and trustworthiness.

Control Domain DG-SEC-004

Monitoring Protection

Purpose:

Detect unauthorized activities.

Control Domain DG-SEC-005

Recovery Protection

Purpose:

Support resilience and restoration.

Control Domain DG-SEC-006

Governance Protection

Purpose:

Ensure accountability and oversight.

7. Classification-Based Security Requirements

PUBLIC

Required Controls:

- Integrity controls
- Availability controls

INTERNAL

Required Controls:

- Access controls
- Monitoring controls

CONFIDENTIAL

Required Controls:

- Role-based access controls
- Encryption in transit
- Encryption at rest
- Audit logging
- Access reviews

RESTRICTED

Required Controls:

- Enhanced access controls
- Encryption controls
- Continuous monitoring
- Security review requirements
- Enhanced audit logging

8. Encryption Governance Requirements

DG-ENC-001

Sensitive data shall be encrypted at rest where required by classification.

DG-ENC-002

Sensitive data shall be encrypted during transmission.

DG-ENC-003

Encryption standards shall be centrally governed.

DG-ENC-004

Encryption key management shall be controlled.

DG-ENC-005

Key access shall be restricted.

DG-ENC-006

Encryption governance activities shall be auditable.

9. Data Integrity Requirements

DG-INTG-001

Critical records shall maintain integrity validation controls.

DG-INTG-002

Lineage records shall be protected from unauthorized modification.

DG-INTG-003

Audit records shall remain immutable.

DG-INTG-004

Replay records shall maintain integrity protections.

DG-INTG-005

Integrity violations shall trigger investigation.

DG-INTG-006

Integrity monitoring records shall be retained.

10. Security Monitoring Requirements

DG-SEC-MON-001

Security-relevant data access shall be monitored.

DG-SEC-MON-002

Privileged activities shall be monitored.

DG-SEC-MON-003

Unauthorized access attempts shall be monitored.

DG-SEC-MON-004

Anomalous activities shall be investigated.

DG-SEC-MON-005

Monitoring events shall be logged.

DG-SEC-MON-006

Monitoring records shall support audit reviews.

11. Incident Governance Requirements

DG-SEC-INC-001

Security incidents shall be documented.

DG-SEC-INC-002

Incident severity shall be classified.

DG-SEC-INC-003

Material incidents shall be escalated.

DG-SEC-INC-004

Root cause analysis shall be performed.

DG-SEC-INC-005

Corrective actions shall be tracked.

DG-SEC-INC-006

Incident records shall be retained.

12. Backup and Recovery Governance

DG-REC-001

Critical data shall be backed up.

DG-REC-002

Backup integrity shall be validated.

DG-REC-003

Recovery procedures shall be documented.

DG-REC-004

Recovery testing shall occur periodically.

DG-REC-005

Recovery activities shall be auditable.

DG-REC-006

Replay-critical datasets shall remain recoverable.

13. Governance Rules

DG-GOV-060

Security controls shall align with classification requirements.

DG-GOV-061

Security monitoring shall remain active.

DG-GOV-062

Unauthorized data access shall be investigated.

DG-GOV-063

Material security incidents shall be escalated.

DG-GOV-064

Security exceptions shall be documented.

DG-GOV-065

Security governance activities shall be auditable.

DG-GOV-066

Protection controls shall support replay and audit requirements.

14. Governance Metrics

DG-MET-066

Encryption Compliance

Target: 100%

DG-MET-067

Security Monitoring Coverage

Target: $\geq 99\%$

DG-MET-068

Critical Backup Coverage

Target: 100%

DG-MET-069

Security Incident Resolution Compliance

Target: $\geq 95\%$

DG-MET-070

Access Control Compliance

Target: $\geq 95\%$

DG-MET-071

Recovery Test Success Rate

Target: $\geq 95\%$

15. Audit Requirements

AUD-DG-066

Security controls shall be auditable.

AUD-DG-067

Encryption governance shall maintain evidence.

AUD-DG-068

Incident investigations shall be reviewable.

AUD-DG-069

Backup and recovery testing shall maintain evidence.

AUD-DG-070

Access monitoring records shall be retained.

AUD-DG-071

Security governance metrics shall be independently validated.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Security Criteria
- SOC 2 Availability Criteria
- SOC 2 Confidentiality Criteria
- ISO 27001 Information Security Controls
- ISO 27001 Asset Protection Controls
- ISO 27002 Security Operations Controls
- Enterprise Procurement Assessments
- Customer Security Reviews

17. Architectural Constraints

AC-DG-064

Security controls shall be implemented independently of business logic processing.

AC-DG-065

Security monitoring systems shall generate immutable audit records.

AC-DG-066

Security governance services shall integrate with Governance, Audit, and Replay modules.

AC-DG-067

Protection controls shall preserve lineage integrity.

AC-DG-068

Recovery mechanisms shall support replay reconstruction requirements.

AC-DG-069

Security governance architecture shall support future compliance expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Unauthorized disclosure
- Data corruption
- Data loss

- Security breaches
- Operational disruption
- Compliance findings

Mitigation Controls:

- Access governance
- Encryption controls
- Continuous monitoring
- Incident response procedures
- Backup and recovery controls
- Audit verification

20. Section Summary

This section establishes the TODP Data Security Governance and Protection Controls Framework, defining security governance responsibilities, protection domains, encryption requirements, integrity controls, monitoring obligations, incident governance processes, recovery requirements, audit obligations, compliance requirements, metrics, and architectural constraints necessary to ensure secure, resilient, auditable, and enterprise-grade protection of all TODP data assets throughout their lifecycle.

DATA GOVERNANCE V1

Section 13 – Data Integration, Exchange & Third-Party Data Governance

Document ID: GOV-DG-013

Document Name: Data Governance V1

Section ID: DG-13

Section Title: Data Integration, Exchange & Third-Party Data Governance

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Integration, Exchange, and Third-Party Data Governance Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines governance controls, ownership responsibilities, approval requirements, validation standards, security obligations, monitoring requirements, audit controls, and compliance expectations governing the movement, exchange, ingestion, export, synchronization, and use of internal and external data sources.

The objective is to ensure that all data integrations and exchanges are secure, traceable, governed, auditable, compliant, and aligned with TODP principles of transparency, explainability, traceability, governance, auditability, and replayability.

2. Objectives

The objectives of this framework are to:

- Establish governance controls for data integrations.
- Ensure trustworthy third-party data usage.
- Support secure data exchange practices.
- Maintain data quality across integration boundaries.
- Preserve lineage and provenance.
- Enable audit and replay requirements.
- Reduce integration-related risks.

- Improve operational transparency.
- Support compliance obligations.
- Maintain enterprise-grade integration governance.

3. Scope

This framework applies to:

Internal Integrations

- Opportunity Services
- Authority Services
- Evidence Services
- Retrieval Services
- Ranking Services
- Recommendation Services
- Governance Services
- Audit Services
- Replay Services

External Integrations

- Data Providers
- Authority Sources
- Evidence Sources
- Identity Services
- Notification Services

- Analytics Services
- Enterprise Customer Integrations
- Future Partner Integrations

The framework applies to:

- APIs
- File Transfers
- Batch Exchanges
- Streaming Exchanges
- Synchronization Processes
- Data Imports
- Data Exports
- Replication Processes

4. Governance Principles

4.1 Trusted Source Principle

External data sources shall be evaluated before operational use.

4.2 Controlled Exchange Principle

Data exchanges shall follow approved governance processes.

4.3 Traceability Principle

All integration activities shall be traceable.

4.4 Security Principle

Data exchanges shall be protected according to classification requirements.

4.5 Accountability Principle

Ownership shall be assigned for every integration.

4.6 Quality Principle

Data quality controls shall be enforced at integration boundaries.

4.7 Auditability Principle

Integration activities shall be auditable.

5. Integration Governance Model

Level 1

Source Governance

Purpose:

Validate source legitimacy and ownership.

Level 2

Exchange Governance

Purpose:

Control data movement and sharing.

Level 3

Quality Governance

Purpose:

Validate incoming and outgoing data quality.

Level 4

Security Governance

Purpose:

Protect exchanged information.

Level 5

Audit Governance

Purpose:

Maintain evidence and traceability.

6. Integration Classification Framework

Category DG-INT-001

Internal System Integration

Examples:

- Service-to-service communication
- Internal APIs
- Internal event processing

Category DG-INT-002

External Data Provider Integration

Examples:

- Evidence providers
- Authority providers
- Data feeds

Category DG-INT-003

Customer Integration

Examples:

- Enterprise customer APIs
- Customer data imports

Category DG-INT-004

Operational Integration

Examples:

- Monitoring systems
- Governance systems
- Audit systems

Category DG-INT-005

Security Integration

Examples:

- Identity providers

- Access management systems
- Security monitoring systems

7. Third-Party Source Governance

DG-TP-001

Third-party sources shall be approved before use.

DG-TP-002

Source ownership shall be documented.

DG-TP-003

Source authority shall be evaluated.

DG-TP-004

Quality assessments shall be performed.

DG-TP-005

Security reviews shall be completed.

DG-TP-006

Source approvals shall be retained.

8. Data Exchange Requirements

DG-EXCH-001

Data exchanges shall be authorized.

DG-EXCH-002

Exchange purposes shall be documented.

DG-EXCH-003

Classification requirements shall be evaluated.

DG-EXCH-004

Exchange activities shall be logged.

DG-EXCH-005

Sensitive information exchanges shall require enhanced controls.

DG-EXCH-006

Exchange records shall be retained.

9. Integration Security Requirements

DG-INT-SEC-001

Authentication controls shall be enforced.

DG-INT-SEC-002

Authorization controls shall be enforced.

DG-INT-SEC-003

Sensitive exchanges shall be encrypted.

DG-INT-SEC-004

Security events shall be monitored.

DG-INT-SEC-005

Unauthorized integration activities shall be investigated.

DG-INT-SEC-006

Security evidence shall be retained.

10. Integration Quality Requirements

DG-INT-QA-001

Inbound data shall be validated.

DG-INT-QA-002

Outbound data shall be validated.

DG-INT-QA-003

Schema compliance shall be verified.

DG-INT-QA-004

Data quality thresholds shall be monitored.

DG-INT-QA-005

Quality exceptions shall be documented.

DG-INT-QA-006

Quality remediation activities shall be tracked.

11. Lineage and Provenance Requirements

DG-INT-LIN-001

External source attribution shall be preserved.

DG-INT-LIN-002

Exchange activities shall maintain lineage records.

DG-INT-LIN-003

Authority sources shall maintain provenance information.

DG-INT-LIN-004

Evidence sources shall maintain provenance information.

DG-INT-LIN-005

Recommendation dependencies shall remain traceable.

DG-INT-LIN-006

Replay activities shall preserve integration lineage.

12. Change Management Requirements

DG-INT-CHG-001

Integration changes shall be reviewed.

DG-INT-CHG-002

Schema modifications shall be approved.

DG-INT-CHG-003

Source changes shall be documented.

DG-INT-CHG-004

Security impacts shall be assessed.

DG-INT-CHG-005

Testing shall be completed before production deployment.

DG-INT-CHG-006

Change records shall be retained.

13. Governance Rules

DG-GOV-067

Unapproved third-party data sources are prohibited.

DG-GOV-068

All integrations shall have assigned ownership.

DG-GOV-069

Data exchanges shall be documented.

DG-GOV-070

Integration activities shall remain auditable.

DG-GOV-071

Lineage and provenance shall be preserved.

DG-GOV-072

Integration exceptions shall be documented.

DG-GOV-073

Integration controls shall support replay and audit requirements.

14. Governance Metrics

DG-MET-072

Approved Integration Coverage

Target: 100%

DG-MET-073

Third-Party Assessment Completion

Target: 100%

DG-MET-074

Integration Quality Compliance

Target: $\geq 98\%$

DG-MET-075

Integration Security Compliance

Target: $\geq 95\%$

DG-MET-076

Lineage Preservation Compliance

Target: $\geq 95\%$

DG-MET-077

Integration Change Approval Compliance

Target: 100%

15. Audit Requirements

AUD-DG-072

Integration inventories shall be auditable.

AUD-DG-073

Third-party approvals shall maintain evidence.

AUD-DG-074

Exchange activities shall be reviewable.

AUD-DG-075

Source assessments shall be retained.

AUD-DG-076

Lineage records shall support reconstruction.

AUD-DG-077

Integration governance metrics shall be independently validated.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Security Criteria
- SOC 2 Processing Integrity Criteria
- ISO 27001 Supplier Relationship Controls
- ISO 27001 Information Transfer Controls
- ISO 27002 Third-Party Governance Controls
- Enterprise Procurement Reviews
- Customer Security Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-070

Integration governance controls shall operate independently from business decision logic.

AC-DG-071

Integration repositories shall maintain historical records.

AC-DG-072

Exchange events shall generate immutable audit records.

AC-DG-073

Integration governance services shall integrate with Governance, Audit, and Replay modules.

AC-DG-074

Provenance information shall remain attached to externally sourced data.

AC-DG-075

Integration architecture shall support future partner ecosystem expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Untrusted data sources
- Data integrity failures

- Unauthorized exchanges
- Security breaches
- Provenance loss
- Audit deficiencies

Mitigation Controls:

- Third-party assessments
- Exchange authorization controls
- Security reviews
- Data quality validation
- Lineage preservation
- Governance oversight

20. Section Summary

This section establishes the TODP Data Integration, Exchange, and Third-Party Data Governance Framework, defining source governance requirements, exchange controls, security obligations, quality standards, lineage requirements, change management processes, audit obligations, compliance requirements, metrics, and architectural constraints necessary to ensure secure, traceable, trustworthy, and enterprise-governed data movement across internal systems and external parties.

DATA GOVERNANCE V1

Section 14 – Data Governance Controls, Policy Exception & Waiver Management

Document ID: GOV-DG-014

Document Name: Data Governance V1

Section ID: DG-14

Section Title: Data Governance Controls, Policy Exception & Waiver Management

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Governance Controls, Policy Exception, and Waiver Management Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines governance control structures, control ownership requirements, exception management procedures, waiver approval processes, risk assessment requirements, monitoring obligations, audit controls, and compliance expectations necessary to ensure controlled deviations from approved governance policies.

The objective is to maintain governance integrity while allowing managed, risk-based exceptions under documented, approved, monitored, and auditable conditions.

2. Objectives

The objectives of this framework are to:

- Establish formal governance control management.
- Define exception and waiver approval processes.
- Ensure accountability for governance deviations.
- Reduce risk associated with policy non-compliance.
- Support audit and compliance obligations.
- Maintain transparency of governance decisions.

- Enable controlled operational flexibility.
- Improve governance oversight.
- Strengthen enterprise risk management.
- Maintain enterprise-grade governance maturity.

3. Scope

This framework applies to:

- Data Classification Controls
- Data Quality Controls
- Metadata Controls
- Lineage Controls
- Retention Controls
- Access Controls
- Privacy Controls
- Security Controls
- Integration Controls
- Audit Controls
- Replay Controls
- Governance Policies
- Governance Standards
- Governance Procedures

The framework applies across all TODP modules:

- Opportunity

- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Governance Principles

4.1 Control Accountability

Every governance control shall have an accountable owner.

4.2 Risk-Based Exceptions

Exceptions shall be evaluated based on risk.

4.3 Temporary Deviation Principle

Exceptions and waivers shall be temporary unless formally renewed.

4.4 Transparency Principle

Exception decisions shall be documented and traceable.

4.5 Auditability Principle

Exception and waiver activities shall be auditable.

4.6 Compensating Controls Principle

Appropriate compensating controls shall be evaluated.

4.7 Governance Oversight Principle

Governance Office shall oversee exception management activities.

5. Governance Control Framework

Control Category DG-CTRL-001

Data Classification Controls

Examples:

- Classification assignment
- Classification review
- Classification enforcement

Control Category DG-CTRL-002

Data Quality Controls

Examples:

- Validation controls
- Quality monitoring
- Quality remediation

Control Category DG-CTRL-003

Metadata Controls

Examples:

- Catalog registration
- Metadata maintenance
- Ownership assignment

Control Category DG-CTRL-004

Lineage Controls

Examples:

- Traceability capture
- Provenance management
- Replay lineage preservation

Control Category DG-CTRL-005

Access Controls

Examples:

- Authorization management
- Access review
- Privileged access monitoring

Control Category DG-CTRL-006

Security Controls

Examples:

- Encryption
- Monitoring
- Incident response

6. Control Ownership Requirements

DG-CTRL-REQ-001

All governance controls shall have assigned owners.

DG-CTRL-REQ-002

Control responsibilities shall be documented.

DG-CTRL-REQ-003

Control effectiveness shall be periodically reviewed.

DG-CTRL-REQ-004

Control inventories shall be maintained.

DG-CTRL-REQ-005

Control ownership changes shall be documented.

DG-CTRL-REQ-006

Ownership records shall be auditable.

7. Exception Management Framework

Definition:

An exception is a temporary approved deviation from a governance requirement.

Exception Categories:

Operational Exception

Purpose:

Address operational constraints.

Technical Exception

Purpose:

Address technology limitations.

Security Exception

Purpose:

Address security control limitations.

Compliance Exception

Purpose:

Address compliance implementation constraints.

Governance Exception

Purpose:

Address governance process constraints.

8. Exception Request Requirements

DG-EXC-001

Exception requests shall be documented.

DG-EXC-002

Business justification shall be provided.

DG-EXC-003

Risk assessment shall be completed.

DG-EXC-004

Compensating controls shall be identified.

DG-EXC-005

Approval authority shall be documented.

DG-EXC-006

Expiration dates shall be defined.

9. Waiver Management Framework

Definition:

A waiver is a formally approved temporary suspension of a governance requirement.

Waiver Characteristics:

- Temporary
- Risk-assessed
- Approved
- Monitored
- Auditable

Waivers shall not permanently replace governance requirements.

10. Approval Authority Model

Low Risk Exception

Approval Authority:

Data Owner

Medium Risk Exception

Approval Authority:

Governance Lead

High Risk Exception

Approval Authority:

Governance Committee

Critical Risk Exception

Approval Authority:

Executive Governance Sponsor

Security-related exceptions may require Security Team approval.

11. Compensating Control Requirements

DG-COMP-001

Compensating controls shall be evaluated.

DG-COMP-002

Control effectiveness shall be documented.

DG-COMP-003

Monitoring requirements shall be defined.

DG-COMP-004

Review schedules shall be established.

DG-COMP-005

Control owners shall be assigned.

DG-COMP-006

Evidence shall be retained.

12. Monitoring and Review Requirements

DG-REV-007

Active exceptions shall be reviewed periodically.

DG-REV-008

Expired exceptions shall be closed or renewed.

DG-REV-009

Waivers shall undergo periodic reassessment.

DG-REV-010

Exception risk ratings shall be reviewed.

DG-REV-011

Monitoring activities shall be documented.

DG-REV-012

Review evidence shall be retained.

13. Governance Rules

DG-GOV-074

Unapproved exceptions are prohibited.

DG-GOV-075

Exceptions shall have defined expiration dates.

DG-GOV-076

Waivers shall undergo formal approval.

DG-GOV-077

Compensating controls shall be evaluated.

DG-GOV-078

Exception activities shall remain auditable.

DG-GOV-079

Governance exception inventories shall be maintained.

DG-GOV-080

Exception management shall support audit and replay requirements.

14. Governance Metrics

DG-MET-078

Approved Exception Documentation Compliance

Target: 100%

DG-MET-079

Exception Review Completion

Target: $\geq 95\%$

DG-MET-080

Expired Exception Closure Compliance

Target: $\geq 95\%$

DG-MET-081

Compensating Control Coverage

Target: $\geq 95\%$

DG-MET-082

Waiver Review Completion

Target: $\geq 95\%$

DG-MET-083

Unauthorized Exception Rate

Target: 0

15. Audit Requirements

AUD-DG-078

Exception inventories shall be auditable.

AUD-DG-079

Approval records shall be retained.

AUD-DG-080

Risk assessments shall be reviewable.

AUD-DG-081

Compensating controls shall maintain evidence.

AUD-DG-082

Exception monitoring activities shall be auditable.

AUD-DG-083

Waiver records shall support independent verification.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Governance Requirements
- SOC 2 Security Criteria
- ISO 27001 Risk Management Controls
- ISO 27001 Exception Management Controls
- ISO 27002 Governance Oversight Controls
- Enterprise Procurement Reviews
- Customer Governance Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-076

Exception repositories shall be logically separated from operational business repositories.

AC-DG-077

Exception records shall support historical retention.

AC-DG-078

Exception approvals shall generate immutable audit records.

AC-DG-079

Exception management services shall integrate with Governance, Audit, and Replay modules.

AC-DG-080

Compensating control records shall remain traceable.

AC-DG-081

Exception management architecture shall support future governance expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Unauthorized policy deviations
- Governance control failures
- Excessive exception accumulation
- Compliance violations
- Audit findings
- Operational governance gaps

Mitigation Controls:

- Formal approval workflows
- Risk assessments
- Compensating controls
- Periodic reviews
- Governance oversight
- Audit verification

20. Section Summary

This section establishes the TODP Data Governance Controls, Policy Exception, and Waiver Management Framework, defining governance control ownership, exception management processes, waiver governance requirements, approval authorities, compensating control obligations, monitoring requirements, audit responsibilities, compliance expectations, metrics, and architectural constraints necessary to ensure controlled, transparent, risk-based, and auditable management of governance deviations across TODP.

DATA GOVERNANCE V1

Section 15 – Data Governance Monitoring, Metrics & Reporting Framework

Document ID: GOV-DG-015

Document Name: Data Governance V1

Section ID: DG-15

Section Title: Data Governance Monitoring, Metrics & Reporting Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Governance Monitoring, Metrics, and Reporting Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines governance performance measurement practices, monitoring controls, reporting requirements, accountability structures, escalation procedures, audit obligations, and compliance expectations necessary to evaluate the effectiveness of the TODP Data Governance Program.

The objective is to provide continuous visibility into governance performance, identify emerging risks, measure control effectiveness, support executive oversight, and maintain enterprise-grade governance maturity across all TODP modules and governed data assets.

2. Objectives

The objectives of this framework are to:

- Measure governance effectiveness.
- Monitor compliance with governance requirements.
- Identify governance risks and control deficiencies.
- Support executive and operational reporting.
- Enable evidence-based governance decisions.
- Improve governance transparency.
- Support audit readiness and compliance obligations.
- Track governance program maturity.
- Drive continuous improvement initiatives.
- Maintain enterprise-scale governance oversight.

3. Scope

This framework applies to:

- Data Ownership Controls
- Classification Controls
- Lifecycle Controls
- Quality Controls
- Metadata Controls
- Lineage Controls
- Retention Controls
- Access Controls
- Privacy Controls
- Security Controls
- Integration Controls
- Exception Controls
- Audit Controls
- Replay Controls

The framework applies across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit

- Replay

4. Governance Monitoring Principles

4.1 Continuous Visibility

Governance performance shall be continuously monitored.

4.2 Measurable Accountability

Governance performance shall be measured using approved metrics.

4.3 Evidence-Based Reporting

Governance reports shall be supported by verifiable evidence.

4.4 Risk Awareness

Monitoring activities shall identify emerging governance risks.

4.5 Transparency

Governance reporting shall support informed decision-making.

4.6 Auditability

Monitoring and reporting activities shall be auditable.

4.7 Continuous Improvement

Metrics shall support governance optimization efforts.

5. Governance Monitoring Framework

Monitoring Layer DG-MON-L1

Operational Monitoring

Purpose:

Track day-to-day governance execution.

Monitoring Layer DG-MON-L2

Control Monitoring

Purpose:

Evaluate governance control effectiveness.

Monitoring Layer DG-MON-L3

Risk Monitoring

Purpose:

Identify governance risks and deficiencies.

Monitoring Layer DG-MON-L4

Compliance Monitoring

Purpose:

Assess policy and regulatory alignment.

Monitoring Layer DG-MON-L5

Executive Monitoring

Purpose:

Provide governance performance visibility to leadership.

6. Governance Metric Categories

Metric Category DG-MCAT-001

Ownership Metrics

Examples:

- Ownership coverage
- Steward assignment coverage

Metric Category DG-MCAT-002

Quality Metrics

Examples:

- Data quality compliance
- Quality incident rates

Metric Category DG-MCAT-003

Access Metrics

Examples:

- Access review completion
- Authorization compliance

Metric Category DG-MCAT-004

Security Metrics

Examples:

- Encryption compliance
- Security incident resolution

Metric Category DG-MCAT-005

Audit Metrics

Examples:

- Audit finding closure
- Control effectiveness

Metric Category DG-MCAT-006

Replay Metrics

Examples:

- Replay success rate
- Replay reconstruction compliance

7. Core Governance KPI Framework

DG-KPI-001

Data Ownership Coverage

Target: 100%

DG-KPI-002

Data Classification Coverage

Target: 100%

DG-KPI-003

Metadata Completeness

Target: $\geq 98\%$

DG-KPI-004

Lineage Coverage

Target: $\geq 95\%$

DG-KPI-005

Retention Compliance

Target: $\geq 95\%$

DG-KPI-006

Access Review Completion

Target: $\geq 95\%$

DG-KPI-007

Governance Exception Compliance

Target: $\geq 95\%$

DG-KPI-008

Audit Finding Closure Rate

Target: $\geq 90\%$

8. Governance Dashboard Requirements

DG-DASH-001

Executive Governance Dashboard

Purpose:

Executive oversight.

DG-DASH-002

Operational Governance Dashboard

Purpose:

Governance operations monitoring.

DG-DASH-003

Data Quality Dashboard

Purpose:

Quality performance monitoring.

DG-DASH-004

Access Governance Dashboard

Purpose:

Authorization monitoring.

DG-DASH-005

Audit Dashboard

Purpose:

Audit performance monitoring.

DG-DASH-006

Replay Governance Dashboard

Purpose:

Replay readiness monitoring.

9. Reporting Framework

Report Type DG-RPT-001

Operational Governance Report

Frequency:

Monthly

Audience:

Governance Office

Report Type DG-RPT-002

Executive Governance Report

Frequency:

Quarterly

Audience:

Executive Governance Sponsor

Report Type DG-RPT-003

Compliance Report

Frequency:

Quarterly

Audience:

Governance Committee

Report Type DG-RPT-004

Audit Readiness Report

Frequency:

Quarterly

Audience:

Audit Team

Report Type DG-RPT-005

Risk Report

Frequency:

Monthly

Audience:

Governance Leadership

10. Threshold Management

Performance Status

Green

Definition:

Within approved target.

Performance Status

Amber

Definition:

Below target but within tolerance.

Performance Status

Red

Definition:

Material deviation requiring escalation.

Threshold definitions shall be documented and approved.

11. Escalation Requirements

DG-ESC-001

Red status metrics shall be escalated.

DG-ESC-002

Repeated control failures shall be escalated.

DG-ESC-003

Material governance risks shall be escalated.

DG-ESC-004

Compliance failures shall be escalated.

DG-ESC-005

Executive reporting shall include unresolved risks.

DG-ESC-006

Escalation records shall be retained.

12. Governance Maturity Assessment

Maturity Level 1

Initial

Maturity Level 2

Managed

Maturity Level 3

Defined

Maturity Level 4

Measured

Maturity Level 5

Optimized

Annual governance maturity assessments shall be performed.

13. Governance Rules

DG-GOV-081

Governance metrics shall be formally defined.

DG-GOV-082

Governance reporting shall be evidence-based.

DG-GOV-083

Metric calculations shall be documented.

DG-GOV-084

Material governance issues shall be reported.

DG-GOV-085

Monitoring records shall be retained.

DG-GOV-086

Governance performance shall undergo periodic review.

DG-GOV-087

Monitoring activities shall support audit and replay requirements.

14. Governance Metrics

DG-MET-084

Dashboard Availability

Target: $\geq 99\%$

DG-MET-085

Reporting Timeliness

Target: $\geq 95\%$

DG-MET-086

Metric Accuracy

Target: $\geq 98\%$

DG-MET-087

Escalation Compliance

Target: $\geq 95\%$

DG-MET-088

Governance Review Completion

Target: $\geq 95\%$

DG-MET-089

Maturity Assessment Completion

Target: 100%

15. Audit Requirements

AUD-DG-084

Metric calculations shall be auditable.

AUD-DG-085

Dashboard data sources shall be traceable.

AUD-DG-086

Governance reports shall maintain evidence.

AUD-DG-087

Escalation records shall be reviewable.

AUD-DG-088

Performance trends shall be retained.

AUD-DG-089

Monitoring controls shall support independent verification.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Governance Requirements
- SOC 2 Monitoring Controls
- ISO 27001 Performance Evaluation Controls
- ISO 27001 Management Review Controls
- ISO 27002 Governance Monitoring Controls
- Enterprise Procurement Reviews
- Customer Governance Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-082

Monitoring services shall operate independently of business decision processing.

AC-DG-083

Metric repositories shall support historical trend analysis.

AC-DG-084

Dashboard systems shall integrate with Governance, Audit, and Replay modules.

AC-DG-085

Reporting systems shall preserve source traceability.

AC-DG-086

Monitoring events shall generate auditable records.

AC-DG-087

Monitoring architecture shall support future governance expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Undetected governance failures
- Inaccurate reporting
- Control degradation
- Compliance deficiencies
- Executive visibility gaps
- Audit findings

Mitigation Controls:

- Continuous monitoring
- KPI management
- Dashboard reporting
- Escalation procedures
- Governance reviews
- Independent audits

20. Section Summary

This section establishes the TODP Data Governance Monitoring, Metrics, and Reporting Framework, defining governance KPIs, monitoring structures, reporting requirements, dashboard standards, escalation procedures, maturity assessment practices, audit obligations, compliance requirements, metrics, and architectural constraints necessary to provide continuous visibility, accountability, and performance management across the TODP Data Governance Program.

DATA GOVERNANCE V1

Section 16 – Data Governance Audit, Assurance & Control Validation Framework

Document ID: GOV-DG-016

Document Name: Data Governance V1

Section ID: DG-16

Section Title: Data Governance Audit, Assurance & Control Validation Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Governance Audit, Assurance, and Control Validation Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the audit governance model, assurance activities, control validation requirements, evidence management practices, accountability structures, reporting obligations, and compliance expectations necessary to verify the effectiveness of the TODP Data Governance Program.

The objective is to provide independent verification that governance controls operate as intended, governance requirements are being followed, risks are appropriately managed, and TODP remains audit-ready for enterprise customers, regulatory reviews, security assessments, and future certification initiatives.

2. Objectives

The objectives of this framework are to:

- Establish governance audit practices.
- Validate effectiveness of governance controls.
- Verify compliance with approved policies.
- Support internal and external audit activities.
- Strengthen governance accountability.
- Improve risk visibility.
- Support enterprise assurance requirements.
- Maintain audit readiness.
- Enable continuous governance improvement.
- Preserve stakeholder trust.

3. Scope

This framework applies to:

- Data Ownership Controls
- Classification Controls
- Lifecycle Controls
- Quality Controls
- Metadata Controls
- Lineage Controls
- Retention Controls
- Access Controls
- Privacy Controls
- Security Controls
- Integration Controls
- Exception Controls
- Monitoring Controls
- Audit Controls
- Replay Controls

The framework applies across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation

- Governance
- Audit
- Replay

4. Assurance Principles

4.1 Independence

Audit and assurance activities shall be performed independently of control ownership.

4.2 Objectivity

Control evaluations shall be evidence-based and unbiased.

4.3 Risk-Based Assessment

Audit priorities shall align with governance risk levels.

4.4 Repeatability

Validation procedures shall be standardized and repeatable.

4.5 Traceability

Assurance conclusions shall be supported by traceable evidence.

4.6 Transparency

Findings and recommendations shall be documented.

4.7 Continuous Improvement

Audit outcomes shall support governance enhancement.

5. Governance Assurance Model

Assurance Layer DG-ASSR-L1

Self-Assessment

Purpose:

Control owner verification of compliance.

Assurance Layer DG-ASSR-L2

Governance Review

Purpose:

Governance Office validation of control operation.

Assurance Layer DG-ASSR-L3

Internal Audit

Purpose:

Independent evaluation of governance effectiveness.

Assurance Layer DG-ASSR-L4

External Assessment

Purpose:

Customer, certification, or third-party reviews.

Assurance Layer DG-ASSR-L5

Executive Oversight

Purpose:

Governance accountability and strategic review.

6. Control Validation Framework

Validation Category DG-VAL-001

Ownership Controls

Examples:

- Ownership assignment
- Steward assignment

Validation Category DG-VAL-002

Classification Controls

Examples:

- Classification coverage
- Classification review compliance

Validation Category DG-VAL-003

Quality Controls

Examples:

- Quality monitoring
- Remediation effectiveness

Validation Category DG-VAL-004

Lineage Controls

Examples:

- Traceability coverage
- Provenance preservation

Validation Category DG-VAL-005

Access Controls

Examples:

- Authorization approvals
- Access review completion

Validation Category DG-VAL-006

Security Controls

Examples:

- Encryption compliance
- Monitoring effectiveness

7. Audit Planning Requirements

DG-AUD-PLAN-001

Annual audit plans shall be developed.

DG-AUD-PLAN-002

Audit scope shall be documented.

DG-AUD-PLAN-003

Risk-based prioritization shall be applied.

DG-AUD-PLAN-004

Audit resources shall be assigned.

DG-AUD-PLAN-005

Stakeholders shall be identified.

DG-AUD-PLAN-006

Audit plans shall be approved.

8. Evidence Management Requirements

DG-EVID-001

Audit evidence shall be collected.

DG-EVID-002

Evidence sources shall be documented.

DG-EVID-003

Evidence integrity shall be maintained.

DG-EVID-004

Evidence repositories shall be secured.

DG-EVID-005

Evidence retention requirements shall be enforced.

DG-EVID-006

Evidence shall support independent verification.

9. Audit Execution Requirements

DG-AUD-EXEC-001

Audit procedures shall be documented.

DG-AUD-EXEC-002

Control testing shall be performed.

DG-AUD-EXEC-003

Evidence shall be evaluated.

DG-AUD-EXEC-004

Exceptions shall be documented.

DG-AUD-EXEC-005

Findings shall be classified.

DG-AUD-EXEC-006

Audit results shall be retained.

10. Finding Classification Framework

Severity Level

Low

Definition:

Limited operational impact.

Severity Level

Medium

Definition:

Moderate governance weakness.

Severity Level

High

Definition:

Significant control deficiency.

Severity Level

Critical

Definition:

Material governance risk requiring executive attention.

Finding classification criteria shall be documented.

11. Remediation Governance

DG-REM-001

Corrective actions shall be assigned.

DG-REM-002

Remediation owners shall be identified.

DG-REM-003

Target completion dates shall be defined.

DG-REM-004

Remediation progress shall be monitored.

DG-REM-005

Closure validation shall be performed.

DG-REM-006

Remediation evidence shall be retained.

12. Assurance Reporting Requirements

DG-RPT-006

Audit Summary Report

Frequency:

Per Audit

DG-RPT-007

Quarterly Assurance Report

Frequency:

Quarterly

DG-RPT-008

Executive Assurance Report

Frequency:

Quarterly

DG-RPT-009

Control Effectiveness Report

Frequency:

Semi-Annual

DG-RPT-010

Audit Readiness Report

Frequency:

Quarterly

13. Governance Rules

DG-GOV-088

Governance controls shall undergo periodic validation.

DG-GOV-089

Audit evidence shall be retained.

DG-GOV-090

Findings shall be formally tracked.

DG-GOV-091

Material deficiencies shall be escalated.

DG-GOV-092

Remediation activities shall be monitored.

DG-GOV-093

Assurance activities shall remain independent.

DG-GOV-094

Audit activities shall support replay and traceability requirements.

14. Governance Metrics

DG-MET-090

Audit Plan Completion

Target: 100%

DG-MET-091

Control Testing Coverage

Target: $\geq 95\%$

DG-MET-092

Finding Closure Rate

Target: $\geq 90\%$

DG-MET-093

Critical Finding Resolution Compliance

Target: $\geq 95\%$

DG-MET-094

Audit Evidence Completeness

Target: $\geq 98\%$

DG-MET-095

Assurance Review Completion

Target: $\geq 95\%$

15. Audit Requirements

AUD-DG-090

Audit procedures shall be documented.

AUD-DG-091

Control testing evidence shall be retained.

AUD-DG-092

Audit findings shall maintain supporting evidence.

AUD-DG-093

Remediation activities shall be independently verifiable.

AUD-DG-094

Assurance reports shall be reviewable.

AUD-DG-095

Audit repositories shall support historical analysis.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Audit Requirements
- SOC 2 Monitoring Controls
- ISO 27001 Internal Audit Controls
- ISO 27001 Performance Evaluation Controls
- ISO 27002 Assurance Practices
- Enterprise Procurement Reviews
- Customer Security Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-088

Audit repositories shall be logically separated from operational repositories.

AC-DG-089

Audit evidence shall support immutability requirements.

AC-DG-090

Assurance systems shall integrate with Governance, Audit, and Replay modules.

AC-DG-091

Control validation records shall support historical retention.

AC-DG-092

Audit reporting systems shall preserve evidence traceability.

AC-DG-093

Assurance architecture shall support future certification initiatives.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Undetected control failures
- Inadequate audit coverage
- Weak remediation practices
- Incomplete evidence
- Governance blind spots
- Compliance deficiencies

Mitigation Controls:

- Risk-based audits
- Independent assurance reviews
- Formal remediation tracking
- Evidence management controls
- Executive oversight
- Continuous validation activities

20. Section Summary

This section establishes the TODP Data Governance Audit, Assurance, and Control Validation Framework, defining audit planning processes, assurance models, control validation requirements, evidence management standards, remediation governance practices, reporting obligations, compliance requirements, metrics, and architectural constraints necessary to ensure independent verification and continuous effectiveness of the TODP Data Governance Program.

DATA GOVERNANCE V1

Section 17 – Data Governance Risk Management Framework

Document ID: GOV-DG-017

Document Name: Data Governance V1

Section ID: DG-17

Section Title: Data Governance Risk Management Framework

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Governance Risk Management Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the governance structure, risk identification processes, assessment methodologies, ownership requirements, treatment strategies, monitoring obligations, reporting mechanisms, audit requirements, and compliance expectations necessary to manage risks associated with TODP data assets and governance operations.

The objective is to proactively identify, assess, manage, monitor, and mitigate governance-related risks that may impact transparency, explainability, traceability, auditability, replayability, security, compliance, operational effectiveness, or enterprise readiness.

2. Objectives

The objectives of this framework are to:

- Establish a formal governance risk management process.
- Identify governance-related risks proactively.
- Support risk-based governance decision making.
- Improve visibility into governance exposure.
- Strengthen accountability for risk ownership.
- Reduce likelihood and impact of governance failures.
- Support audit and compliance requirements.
- Enable executive risk oversight.
- Improve governance resilience.
- Maintain enterprise-grade risk governance maturity.

3. Scope

This framework applies to risks affecting:

- Opportunity Data
- Authority Data
- Evidence Data
- Retrieval Data
- Ranking Data
- Recommendation Data
- Governance Records
- Audit Records
- Replay Records
- Metadata Repositories
- Access Governance
- Privacy Governance
- Security Governance
- Integration Governance
- Retention Governance

The framework applies across:

- Operational Activities
- Governance Activities
- Audit Activities
- Replay Activities
- Technology Services
- Third-Party Integrations

4. Risk Management Principles

4.1 Risk Ownership

All governance risks shall have assigned owners.

4.2 Risk-Based Decision Making

Governance decisions shall consider risk impact and likelihood.

4.3 Early Detection

Risk identification shall occur proactively.

4.4 Continuous Monitoring

Risks shall be continuously monitored.

4.5 Transparency

Risk reporting shall be accurate and timely.

4.6 Accountability

Risk treatment activities shall be assigned and tracked.

4.7 Auditability

Risk management activities shall be auditable.

5. Governance Risk Categories

Risk Category DG-RISK-001

Data Quality Risk

Examples:

- Inaccurate records
- Incomplete data
- Quality degradation

Risk Category DG-RISK-002

Data Security Risk

Examples:

- Unauthorized access
- Data disclosure
- Data corruption

Risk Category DG-RISK-003

Privacy Risk

Examples:

- Improper information handling
- Sensitive data exposure

Risk Category DG-RISK-004

Compliance Risk

Examples:

- Policy violations
- Audit deficiencies

Risk Category DG-RISK-005

Lineage Risk

Examples:

- Traceability loss
- Provenance failures

Risk Category DG-RISK-006

Replay Risk

Examples:

- Replay reconstruction failures
- Historical inconsistency

Risk Category DG-RISK-007

Third-Party Risk

Examples:

- External data quality issues
- Supplier governance weaknesses

6. Risk Governance Structure

Executive Governance Sponsor

Responsibilities:

- Risk oversight
- Risk acceptance authority

Governance Committee

Responsibilities:

- Risk review
- Risk prioritization

Governance Office

Responsibilities:

- Framework administration
- Risk reporting

Risk Owners

Responsibilities:

- Risk management
- Treatment execution

Audit Team

Responsibilities:

- Independent validation

7. Risk Identification Requirements

DG-RISK-ID-001

Risks shall be identified periodically.

DG-RISK-ID-002

Material governance changes shall trigger risk reviews.

DG-RISK-ID-003

Audit findings shall be evaluated for risk implications.

DG-RISK-ID-004

Security incidents shall be evaluated for governance risks.

DG-RISK-ID-005

Third-party changes shall be assessed.

DG-RISK-ID-006

Risk identification records shall be retained.

8. Risk Assessment Framework

Likelihood Ratings

1 – Rare

2 – Unlikely

3 – Possible

4 – Likely

5 – Almost Certain

Impact Ratings

1 – Negligible

2 – Minor

3 – Moderate

4 – Major

5 – Severe

Risk Score

Risk Score = Likelihood × Impact

Assessment results shall be documented.

9. Risk Classification Framework

Low Risk

Score Range:

1 – 4

Medium Risk

Score Range:

5 – 9

High Risk

Score Range:

10 – 16

Critical Risk

Score Range:

17 – 25

Risk treatment requirements shall align with classification levels.

10. Risk Treatment Strategies

Strategy DG-TREAT-001

Mitigate

Purpose:

Reduce likelihood or impact.

Strategy DG-TREAT-002

Transfer

Purpose:

Shift responsibility through contractual or operational arrangements.

Strategy DG-TREAT-003

Avoid

Purpose:

Eliminate risk-generating activities.

Strategy DG-TREAT-004

Accept

Purpose:

Formally acknowledge and monitor residual risk.

Acceptance authority shall align with risk severity.

11. Risk Register Requirements

DG-RR-001

All material risks shall be recorded.

DG-RR-002

Risk owners shall be assigned.

DG-RR-003

Treatment plans shall be documented.

DG-RR-004

Target completion dates shall be defined.

DG-RR-005

Risk status shall be tracked.

DG-RR-006

Historical records shall be retained.

12. Risk Monitoring Requirements

DG-RISK-MON-001

High and Critical risks shall be reviewed regularly.

DG-RISK-MON-002

Treatment progress shall be monitored.

DG-RISK-MON-003

Emerging risks shall be evaluated.

DG-RISK-MON-004

Residual risks shall be reassessed.

DG-RISK-MON-005

Monitoring activities shall be documented.

DG-RISK-MON-006

Risk metrics shall be maintained.

13. Governance Rules

DG-GOV-095

Material risks shall be documented.

DG-GOV-096

Risk ownership shall be assigned.

DG-GOV-097

Critical risks shall be escalated.

DG-GOV-098

Risk treatment plans shall be tracked.

DG-GOV-099

Risk acceptance shall be approved.

DG-GOV-100

Risk records shall remain auditable.

DG-GOV-101

Risk management activities shall support audit and replay requirements.

14. Governance Metrics

DG-MET-096

Risk Register Coverage

Target: 100%

DG-MET-097

Risk Assessment Completion

Target: $\geq 95\%$

DG-MET-098

High Risk Treatment Completion

Target: $\geq 90\%$

DG-MET-099

Critical Risk Review Compliance

Target: 100%

DG-MET-100

Residual Risk Review Completion

Target: $\geq 95\%$

DG-MET-101

Risk Ownership Assignment Coverage

Target: 100%

15. Audit Requirements

AUD-DG-096

Risk assessments shall be auditable.

AUD-DG-097

Risk treatment plans shall maintain evidence.

AUD-DG-098

Risk acceptance decisions shall be documented.

AUD-DG-099

Risk monitoring activities shall be reviewable.

AUD-DG-100

Risk registers shall support independent validation.

AUD-DG-101

Risk governance metrics shall be verifiable.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Risk Assessment Criteria

- SOC 2 Governance Requirements
- ISO 27001 Risk Management Controls
- ISO 27001 Governance Controls
- ISO 27002 Risk Assessment Practices
- Enterprise Procurement Reviews
- Customer Governance Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-094

Risk repositories shall be logically separated from operational processing repositories.

AC-DG-095

Risk records shall support historical versioning.

AC-DG-096

Risk management services shall integrate with Governance, Audit, and Replay modules.

AC-DG-097

Risk assessment records shall maintain traceability.

AC-DG-098

Risk reporting systems shall support executive oversight requirements.

AC-DG-099

Risk governance architecture shall support future regulatory and certification requirements.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Governance control failures
- Data quality degradation
- Security incidents
- Privacy violations
- Traceability breakdowns
- Replay failures

Mitigation Controls:

- Formal risk assessments
- Risk ownership assignments
- Continuous monitoring
- Governance reviews
- Audit validation
- Executive oversight

20. Section Summary

This section establishes the TODP Data Governance Risk Management Framework, defining governance risk categories, assessment methodologies, ownership responsibilities, treatment strategies, monitoring requirements, reporting obligations, audit controls, compliance expectations, metrics, and architectural constraints necessary to proactively identify, manage, and mitigate governance risks across all TODP data assets and governance operations.

DATA GOVERNANCE V1

Section 18 – Data Governance Operating Model & Organizational Structure

Document ID: GOV-DG-018

Document Name: Data Governance V1

Section ID: DG-18

Section Title: Data Governance Operating Model & Organizational Structure

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Governance Operating Model and Organizational Structure for the Transparent Opportunity Decision Platform (TODP).

The operating model defines governance authorities, organizational responsibilities, decision-making structures, accountability mechanisms, operating procedures, escalation pathways, and collaboration frameworks required to execute the TODP Data Governance Program.

The objective is to ensure governance activities are performed consistently, transparently, efficiently, and accountably across all TODP modules while supporting enterprise scalability, audit readiness, compliance obligations, operational effectiveness, and future organizational growth.

2. Objectives

The objectives of this operating model are to:

- Establish governance organizational structures.
- Define governance roles and responsibilities.
- Clarify decision-making authority.
- Improve governance accountability.
- Support cross-functional collaboration.
- Enable governance scalability.
- Support audit and compliance activities.
- Improve governance execution consistency.
- Reduce organizational ambiguity.
- Maintain enterprise-grade governance operations.

3. Scope

This operating model applies to:

- Data Governance Activities
- Data Ownership Activities
- Data Stewardship Activities
- Metadata Governance
- Lineage Governance
- Quality Governance
- Access Governance

- Privacy Governance
- Security Governance
- Integration Governance
- Audit Governance
- Replay Governance

The model applies across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Operating Principles

4.1 Accountability

Governance responsibilities shall be clearly assigned.

4.2 Transparency

Governance decisions shall be documented and traceable.

4.3 Collaboration

Governance activities shall support cross-functional participation.

4.4 Scalability

The operating model shall support organizational growth.

4.5 Independence

Audit and assurance functions shall remain independent.

4.6 Consistency

Governance processes shall be standardized.

4.7 Continuous Improvement

Governance operations shall evolve through measurable improvement initiatives.

5. Governance Organizational Structure

Level 1

Executive Governance Sponsor

Responsibilities:

- Strategic oversight
- Governance sponsorship
- Risk acceptance
- Executive escalation

Level 2

Governance Committee

Responsibilities:

- Governance oversight
- Policy approval
- Cross-domain decisions
- Exception approvals

Level 3

Governance Office

Responsibilities:

- Program administration
- Governance coordination
- Metrics reporting

- Governance monitoring

Level 4

Domain Governance Leads

Responsibilities:

- Domain governance execution
- Stewardship coordination
- Control oversight

Level 5

Operational Governance Participants

Responsibilities:

- Daily governance activities
- Compliance execution
- Issue management

6. Governance Committee Structure

Committee ID:

DG-COM-001

Purpose:

Provide enterprise governance oversight.

Membership:

- Executive Governance Sponsor
- Governance Lead
- Security Lead
- Audit Representative
- Domain Governance Leads

Responsibilities:

- Governance strategy
- Policy approval
- Risk review
- Escalation resolution
- Program oversight

Meeting Frequency:

Quarterly or as required.

7. Governance Office Responsibilities

DG-OFF-001

Maintain governance policies.

DG-OFF-002

Coordinate governance reviews.

DG-OFF-003

Maintain governance metrics.

DG-OFF-004

Manage governance reporting.

DG-OFF-005

Support audits and assessments.

DG-OFF-006

Coordinate continuous improvement initiatives.

8. Domain Governance Structure

Domain DG-DOM-OPP

Opportunity Governance

Domain DG-DOM-AUTH

Authority Governance

Domain DG-DOM-EVD

Evidence Governance

Domain DG-DOM-RET

Retrieval Governance

Domain DG-DOM-RANK

Ranking Governance

Domain DG-DOM-REC

Recommendation Governance

Domain DG-DOM-GOV

Governance Operations

Domain DG-DOM-AUD

Audit Governance

Domain DG-DOM-REP

Replay Governance

Each domain shall maintain assigned ownership and stewardship.

9. Decision Authority Model

Decision Category

Operational Governance

Authority:

Domain Governance Lead

Decision Category

Cross-Domain Governance

Authority:

Governance Committee

Decision Category

Risk Acceptance

Authority:

Executive Governance Sponsor

Decision Category

Policy Approval

Authority:

Governance Committee

Decision Category

Exception Approval

Authority:

Assigned Approval Authority Based on Risk Level

10. Governance Escalation Framework

Level 1

Data Steward

Issue Types:

- Operational governance issues

Level 2

Data Owner

Issue Types:

- Domain governance issues

Level 3

Governance Lead

Issue Types:

- Cross-domain governance issues

Level 4

Governance Committee

Issue Types:

- Material governance conflicts

Level 5

Executive Governance Sponsor

Issue Types:

- Strategic or critical governance matters

11. Governance Meeting Framework

Meeting Type

Operational Governance Review

Frequency:

Monthly

Meeting Type

Domain Governance Review

Frequency:

Monthly

Meeting Type

Governance Committee Review

Frequency:

Quarterly

Meeting Type

Executive Governance Review

Frequency:

Quarterly

Meeting Type

Audit Readiness Review

Frequency:

Quarterly

12. RACI Governance Model

Policy Development

Responsible:

Governance Office

Accountable:

Governance Committee

Consulted:

Security Team, Audit Team

Informed:

Domain Leads

Ownership Assignment

Responsible:

Data Steward

Accountable:

Data Owner

Consulted:

Governance Office

Informed:

Audit Team

Risk Management

Responsible:

Risk Owner

Accountable:

Governance Lead

Consulted:

Governance Committee

Informed:

Executive Sponsor

Audit Review

Responsible:

Audit Team

Accountable:

Audit Lead

Consulted:

Governance Office

Informed:

Executive Sponsor

13. Governance Rules

DG-GOV-102

Governance responsibilities shall be documented.

DG-GOV-103

Governance decisions shall be traceable.

DG-GOV-104

Governance meetings shall maintain records.

DG-GOV-105

Escalation procedures shall be followed.

DG-GOV-106

Cross-domain conflicts shall be formally resolved.

DG-GOV-107

Governance activities shall be auditable.

DG-GOV-108

Operating model activities shall support audit and replay requirements.

14. Governance Metrics

DG-MET-102

Governance Meeting Completion

Target: $\geq 95\%$

DG-MET-103

Governance Review Completion

Target: $\geq 95\%$

DG-MET-104

Escalation Resolution Compliance

Target: $\geq 95\%$

DG-MET-105

Ownership Assignment Coverage

Target: 100%

DG-MET-106

Governance Decision Documentation Compliance

Target: 100%

DG-MET-107

Governance Committee Attendance

Target: $\geq 90\%$

15. Audit Requirements

AUD-DG-102

Governance decisions shall be auditable.

AUD-DG-103

Committee records shall be retained.

AUD-DG-104

Meeting minutes shall be reviewable.

AUD-DG-105

Escalation records shall maintain evidence.

AUD-DG-106

Decision authorities shall be verifiable.

AUD-DG-107

Governance reporting shall support independent review.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Governance Requirements
- SOC 2 Oversight Controls
- ISO 27001 Leadership Controls
- ISO 27001 Organizational Governance Controls
- ISO 27002 Governance Management Practices
- Enterprise Procurement Reviews
- Customer Governance Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-100

Governance repositories shall preserve decision history.

AC-DG-101

Governance workflows shall support approval traceability.

AC-DG-102

Operating model records shall integrate with Governance, Audit, and Replay modules.

AC-DG-103

Decision records shall support historical reconstruction.

AC-DG-104

Governance reporting services shall preserve source traceability.

AC-DG-105

Operating model architecture shall support future organizational expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Unclear accountability
- Governance decision delays
- Escalation failures
- Cross-domain conflicts
- Organizational inefficiencies

- Audit findings

Mitigation Controls:

- Defined governance roles
- Formal escalation procedures
- Governance committee oversight
- RACI structures
- Documentation controls
- Audit validation

20. Section Summary

This section establishes the TODP Data Governance Operating Model and Organizational Structure, defining governance authorities, committee structures, operating responsibilities, decision-making processes, escalation pathways, meeting frameworks, accountability mechanisms, audit obligations, compliance requirements, metrics, and architectural constraints necessary to ensure effective, scalable, transparent, and enterprise-grade execution of the TODP Data Governance Program.

DATA GOVERNANCE V1

Section 19 – Data Governance Maturity, Continuous Improvement & Program Evolution Framework

Document ID: GOV-DG-019

Document Name: Data Governance V1

Section ID: DG-19

Section Title: Data Governance Maturity, Continuous Improvement & Program Evolution Framework

Version: 1.0

Status: Approved

Owner: Governance Office

1. Purpose

This section establishes the Data Governance Maturity, Continuous Improvement, and Program Evolution Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines the methodology, governance structures, assessment models, improvement mechanisms, accountability requirements, reporting obligations, and strategic planning activities necessary to continuously enhance the effectiveness, efficiency, scalability, and resilience of the TODP Data Governance Program.

The objective is to ensure that governance capabilities evolve alongside business growth, platform expansion, customer requirements, regulatory expectations, security obligations, and enterprise maturity objectives.

2. Objectives

The objectives of this framework are to:

- Establish a governance maturity model.
- Measure governance program effectiveness.
- Identify improvement opportunities.
- Support strategic governance planning.
- Drive continuous governance optimization.
- Improve operational efficiency.
- Enhance audit and compliance readiness.
- Support organizational scalability.
- Reduce governance-related risks.
- Maintain enterprise-grade governance excellence.

3. Scope

This framework applies to:

- Governance Policies
- Governance Processes
- Governance Controls
- Governance Metrics
- Governance Reporting
- Governance Technology
- Governance Operations
- Governance Oversight
- Audit Readiness
- Compliance Readiness
- Risk Management
- Replay Governance

The framework applies across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance

- Audit
- Replay

4. Continuous Improvement Principles

4.1 Measurable Progress

Governance improvement shall be supported by measurable outcomes.

4.2 Risk-Based Prioritization

Improvement activities shall prioritize governance risks and business impact.

4.3 Evidence-Based Decisions

Program evolution decisions shall rely on verifiable evidence.

4.4 Stakeholder Alignment

Improvement initiatives shall support business and governance objectives.

4.5 Scalability

Governance improvements shall support future growth.

4.6 Transparency

Improvement activities shall be documented and reported.

4.7 Continuous Learning

Lessons learned shall drive governance evolution.

5. Governance Maturity Model

Maturity Level DG-MAT-001

Initial

Characteristics:

- Ad hoc governance activities
- Limited standardization
- Reactive controls

Maturity Level DG-MAT-002

Managed

Characteristics:

- Defined governance ownership
- Basic governance controls
- Documented procedures

Maturity Level DG-MAT-003

Defined

Characteristics:

- Standardized governance processes
- Established metrics
- Consistent governance execution

Maturity Level DG-MAT-004

Measured

Characteristics:

- Quantitative governance monitoring
- Risk-driven governance management
- Formal reporting

Maturity Level DG-MAT-005

Optimized

Characteristics:

- Continuous improvement culture
- Predictive governance insights
- Automated governance capabilities

Target TODP Maturity State:

Level 4 (Measured) progressing toward Level 5 (Optimized).

6. Governance Assessment Framework

Assessment Domain DG-ASMT-001

Ownership & Accountability

Assessment Domain DG-ASMT-002

Classification Governance

Assessment Domain DG-ASMT-003

Quality Governance

Assessment Domain DG-ASMT-004

Metadata Governance

Assessment Domain DG-ASMT-005

Lineage Governance

Assessment Domain DG-ASMT-006

Retention Governance

Assessment Domain DG-ASMT-007

Access Governance

Assessment Domain DG-ASMT-008

Privacy Governance

Assessment Domain DG-ASMT-009

Security Governance

Assessment Domain DG-ASMT-010

Audit & Replay Governance

7. Assessment Requirements

DG-ASMT-REQ-001

Formal maturity assessments shall occur annually.

DG-ASMT-REQ-002

Assessment criteria shall be documented.

DG-ASMT-REQ-003

Assessment results shall be retained.

DG-ASMT-REQ-004

Improvement opportunities shall be identified.

DG-ASMT-REQ-005

Assessment findings shall be reported.

DG-ASMT-REQ-006

Assessment records shall be auditable.

8. Continuous Improvement Framework

Improvement Stage 1

Assessment

Purpose:

Identify governance gaps.

Improvement Stage 2

Prioritization

Purpose:

Rank opportunities by risk and business value.

Improvement Stage 3

Planning

Purpose:

Develop improvement roadmap.

Improvement Stage 4

Execution

Purpose:

Implement approved initiatives.

Improvement Stage 5

Validation

Purpose:

Verify effectiveness.

Improvement Stage 6

Institutionalization

Purpose:

Integrate improvements into standard governance operations.

9. Governance Improvement Sources

Improvement opportunities may originate from:

- Governance Assessments
- Audit Findings
- Risk Assessments
- Compliance Reviews
- Customer Assessments

- Security Reviews
- Incident Investigations
- Replay Validation Results
- Executive Reviews
- Operational Feedback

All improvement sources shall be documented.

10. Governance Roadmap Management

DG-RMAP-001

Improvement roadmaps shall be maintained.

DG-RMAP-002

Roadmaps shall align with strategic objectives.

DG-RMAP-003

Roadmaps shall identify accountable owners.

DG-RMAP-004

Target completion dates shall be established.

DG-RMAP-005

Progress shall be monitored.

DG-RMAP-006

Roadmap updates shall be documented.

11. Lessons Learned Framework

DG-LESSON-001

Governance incidents shall undergo review.

DG-LESSON-002

Audit findings shall be analyzed.

DG-LESSON-003

Control failures shall be evaluated.

DG-LESSON-004

Improvement recommendations shall be documented.

DG-LESSON-005

Knowledge shall be shared with stakeholders.

DG-LESSON-006

Lessons learned records shall be retained.

12. Program Evolution Requirements

DG-EVOL-001

Governance policies shall be periodically reviewed.

DG-EVOL-002

Governance standards shall be updated as needed.

DG-EVOL-003

Emerging risks shall be evaluated.

DG-EVOL-004

Technology changes shall be assessed.

DG-EVOL-005

Regulatory developments shall be monitored.

DG-EVOL-006

Program evolution decisions shall be documented.

13. Governance Rules

DG-GOV-109

Maturity assessments shall be performed.

DG-GOV-110

Improvement plans shall be documented.

DG-GOV-111

Improvement initiatives shall be tracked.

DG-GOV-112

Lessons learned shall be retained.

DG-GOV-113

Program evolution activities shall be evidence-based.

DG-GOV-114

Governance effectiveness shall be periodically reviewed.

DG-GOV-115

Continuous improvement activities shall support audit and replay requirements.

14. Governance Metrics

DG-MET-108

Maturity Assessment Completion

Target: 100%

DG-MET-109

Improvement Initiative Completion

Target: $\geq 90\%$

DG-MET-110

Roadmap Progress Compliance

Target: $\geq 90\%$

DG-MET-111

Lessons Learned Review Completion

Target: $\geq 95\%$

DG-MET-112

Policy Review Completion

Target: $\geq 95\%$

DG-MET-113

Governance Capability Improvement Rate

Target: Positive Year-over-Year Improvement

15. Audit Requirements

AUD-DG-108

Maturity assessments shall be auditable.

AUD-DG-109

Improvement roadmaps shall maintain evidence.

AUD-DG-110

Program evolution decisions shall be reviewable.

AUD-DG-111

Lessons learned repositories shall be retained.

AUD-DG-112

Improvement initiative tracking shall be verifiable.

AUD-DG-113

Governance maturity metrics shall support independent validation.

16. Compliance Requirements

This framework supports alignment with:

- SOC 2 Continuous Monitoring Requirements
- SOC 2 Governance Requirements
- ISO 27001 Continual Improvement Controls
- ISO 27001 Management Review Controls
- ISO 27002 Governance Improvement Practices

- Enterprise Procurement Reviews
- Customer Governance Assessments
- Internal Audit Programs

17. Architectural Constraints

AC-DG-106

Improvement repositories shall preserve historical records.

AC-DG-107

Assessment records shall support version history.

AC-DG-108

Improvement tracking systems shall integrate with Governance, Audit, and Replay modules.

AC-DG-109

Program evolution records shall preserve traceability.

AC-DG-110

Governance reporting systems shall support maturity trend analysis.

AC-DG-111

Improvement architecture shall support future organizational growth.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Governance stagnation
- Control degradation
- Unaddressed audit findings
- Regulatory misalignment
- Organizational scaling challenges
- Maturity plateau

Mitigation Controls:

- Annual maturity assessments
- Improvement roadmaps
- Lessons learned programs
- Executive oversight
- Continuous monitoring
- Governance reviews

20. Section Summary

This section establishes the TODP Data Governance Maturity, Continuous Improvement, and Program Evolution Framework, defining maturity assessment methodologies, improvement processes, roadmap management practices, lessons learned programs, evolution requirements, governance rules, audit obligations, compliance requirements, metrics, and architectural constraints

necessary to ensure the continuous advancement and long-term effectiveness of the TODP Data Governance Program.

DATA GOVERNANCE V1

Section 20 – Data Governance Compliance Mapping, Regulatory Alignment & Document Control

Document ID: GOV-DG-020

Document Name: Data Governance V1

Section ID: DG-20

Section Title: Data Governance Compliance Mapping, Regulatory Alignment & Document Control

Version: 1.0

Status: Approved

Owner: Governance Office

Classification: Internal Confidential

1. Purpose

This section establishes the Data Governance Compliance Mapping, Regulatory Alignment, and Document Control Framework for the Transparent Opportunity Decision Platform (TODP).

The framework defines compliance alignment practices, regulatory mapping requirements, governance documentation controls, document lifecycle procedures, review responsibilities, audit obligations, and maintenance requirements necessary to ensure that the TODP Data Governance Program remains controlled, current, traceable, auditable, and aligned with enterprise governance expectations.

The objective is to provide a structured mechanism for demonstrating governance compliance, maintaining governance documentation integrity, supporting customer due diligence reviews, enabling audit readiness, and preparing for future certification and regulatory requirements.

2. Objectives

The objectives of this framework are to:

- Establish compliance mapping practices.
- Maintain regulatory alignment visibility.
- Govern documentation lifecycle activities.
- Support audit and assurance activities.
- Demonstrate governance control coverage.
- Improve customer and stakeholder confidence.
- Enable efficient governance reviews.
- Maintain traceable document ownership.
- Support future certification initiatives.
- Ensure long-term governance sustainability.

3. Scope

This framework applies to:

- Data Governance Policies
- Data Governance Standards
- Data Governance Procedures
- Governance Framework Documents
- Governance Metrics Definitions
- Compliance Mappings
- Audit Evidence References
- Regulatory Alignment Records
- Governance Reports
- Governance Registers

The framework applies across:

- Opportunity
- Authority
- Evidence
- Retrieval
- Ranking
- Recommendation
- Governance
- Audit
- Replay

4. Compliance Governance Principles

4.1 Traceable Compliance

Compliance obligations shall be mapped to governance controls.

4.2 Evidence-Based Alignment

Compliance assertions shall be supported by documented evidence.

4.3 Continuous Maintenance

Compliance mappings shall be periodically reviewed.

4.4 Ownership Accountability

Compliance artifacts shall have assigned owners.

4.5 Auditability

Compliance activities shall be auditable.

4.6 Version Control

Governance documentation shall maintain controlled versions.

4.7 Regulatory Awareness

Emerging requirements shall be monitored and evaluated.

5. Compliance Mapping Framework

Compliance mappings shall establish relationships between:

- Governance Requirements
- Governance Controls
- Policies
- Standards
- Procedures
- Audit Evidence
- Metrics

- Risks
- Corrective Actions

Mappings shall support traceability from requirement to evidence.

6. Regulatory Alignment Domains

Alignment Domain DG-COMP-001

SOC 2 Readiness

Coverage Areas:

- Security
- Availability
- Confidentiality
- Processing Integrity
- Privacy

Alignment Domain DG-COMP-002

ISO 27001 Readiness

Coverage Areas:

- Governance
- Risk Management

- Asset Management
- Access Control
- Operations Security

Alignment Domain DG-COMP-003

ISO 27002 Alignment

Coverage Areas:

- Governance Practices
- Information Controls
- Operational Controls

Alignment Domain DG-COMP-004

Enterprise Procurement Readiness

Coverage Areas:

- Governance
- Auditability
- Security
- Compliance

Alignment Domain DG-COMP-005

Customer Assurance Readiness

Coverage Areas:

- Governance Transparency
- Control Evidence
- Operational Maturity

7. Compliance Inventory Requirements

DG-CI-001

Compliance obligations shall be documented.

DG-CI-002

Applicable controls shall be identified.

DG-CI-003

Control owners shall be assigned.

DG-CI-004

Evidence sources shall be documented.

DG-CI-005

Review schedules shall be established.

DG-CI-006

Inventory records shall be retained.

8. Control Mapping Requirements

DG-MAP-001

Governance controls shall be mapped to compliance obligations.

DG-MAP-002

Control gaps shall be documented.

DG-MAP-003

Compensating controls shall be identified where applicable.

DG-MAP-004

Mapping changes shall be reviewed.

DG-MAP-005

Mapping records shall be version controlled.

DG-MAP-006

Mappings shall support audit verification.

9. Document Governance Framework

Document Category DG-DOC-001

Policies

Document Category DG-DOC-002

Standards

Document Category DG-DOC-003

Procedures

Document Category DG-DOC-004

Guidelines

Document Category DG-DOC-005

Registers

Document Category DG-DOC-006

Governance Reports

All governed documents shall follow approved document control processes.

10. Document Lifecycle Management

Stage 1

Draft

Stage 2

Review

Stage 3

Approval

Stage 4

Publication

Stage 5

Maintenance

Stage 6

Retirement

Each lifecycle stage shall be documented and auditable.

11. Document Control Requirements

DG-DOC-CTRL-001

Documents shall have assigned owners.

DG-DOC-CTRL-002

Documents shall maintain version history.

DG-DOC-CTRL-003

Document approvals shall be recorded.

DG-DOC-CTRL-004

Document reviews shall occur periodically.

DG-DOC-CTRL-005

Retired documents shall be archived.

DG-DOC-CTRL-006

Document repositories shall be governed.

12. Review and Maintenance Requirements

DG-REV-013

Governance policies shall be reviewed annually.

DG-REV-014

Compliance mappings shall be reviewed annually.

DG-REV-015

Material governance changes shall trigger reviews.

DG-REV-016

Regulatory developments shall be assessed.

DG-REV-017

Review evidence shall be retained.

DG-REV-018

Maintenance activities shall be documented.

13. Governance Rules

DG-GOV-116

Compliance mappings shall be maintained.

DG-GOV-117

Governance documents shall be version controlled.

DG-GOV-118

Document ownership shall be assigned.

DG-GOV-119

Governance reviews shall be documented.

DG-GOV-120

Regulatory alignment assessments shall be retained.

DG-GOV-121

Compliance evidence shall be traceable.

DG-GOV-122

Document control activities shall support audit and replay requirements.

14. Governance Metrics

DG-MET-114

Policy Review Completion

Target: $\geq 95\%$

DG-MET-115

Compliance Mapping Coverage

Target: 100%

DG-MET-116

Document Ownership Coverage

Target: 100%

DG-MET-117

Version Control Compliance

Target: 100%

DG-MET-118

Regulatory Review Completion

Target: $\geq 95\%$

DG-MET-119

Document Approval Compliance

Target: 100%

15. Audit Requirements

AUD-DG-114

Compliance mappings shall be auditable.

AUD-DG-115

Control coverage evidence shall be reviewable.

AUD-DG-116

Document approvals shall be retained.

AUD-DG-117

Version histories shall be maintained.

AUD-DG-118

Review activities shall maintain evidence.

AUD-DG-119

Regulatory alignment assessments shall support independent verification.

16. Compliance Requirements

This framework directly supports:

- SOC 2 Readiness Programs
- ISO 27001 Readiness Programs
- ISO 27002 Alignment Activities
- Enterprise Procurement Reviews

- Customer Security Assessments
- Governance Due Diligence Reviews
- Internal Audit Programs
- Future Certification Initiatives

17. Architectural Constraints

AC-DG-112

Governance document repositories shall support version control.

AC-DG-113

Compliance mappings shall preserve traceability relationships.

AC-DG-114

Document control systems shall integrate with Governance, Audit, and Replay modules.

AC-DG-115

Governance repositories shall preserve historical records.

AC-DG-116

Compliance reporting services shall support evidence traceability.

AC-DG-117

Document management architecture shall support future governance expansion.

18. Traceability

Related Documents

BRD V1

SRS V1

SAD V1

TDD V1

TRM V1

TEST SPEC V1

RTM V1

ADR V1

API SPEC V1

Mapped Modules

Opportunity

Authority

Evidence

Retrieval

Ranking

Recommendation

Governance

Audit

Replay

19. Risk Considerations

Primary Risks:

- Compliance gaps
- Outdated governance documentation
- Missing control mappings
- Audit deficiencies
- Regulatory misalignment
- Evidence traceability failures

Mitigation Controls:

- Compliance inventories
- Annual reviews
- Version control processes
- Document ownership assignments
- Audit verification
- Governance oversight

20. Section Summary

This section establishes the TODP Data Governance Compliance Mapping, Regulatory Alignment, and Document Control Framework, defining compliance mapping practices, regulatory alignment requirements, document lifecycle controls, ownership responsibilities, review obligations, audit

requirements, compliance expectations, metrics, and architectural constraints necessary to maintain a controlled, traceable, auditable, and enterprise-ready Data Governance Program across TODP.